

Looking Ahead to Act Together: An Integrative Review and Theoretical Framework of Future Cognition and Long- Term Collective Action

Supplementary Materials

0707/2026
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Guide to Supplementary Materials

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Data Availability

All data and scripts for this project are available:
<https://github.com/ksyme/future-cognition-collective-action-integrative-review>

Project Page and Preregistration

<https://osf.io/p9mgt>

Brief description

- Platform: OSF
- Template: Van den Akker, O. R., Peters, G. Y., Bakker, C., Carlsson, R., Coles, N. A., Corker, K. S., Feldman, G., , Moreau, D., Nordström, T., Pickering, J. S., Riegelman, A., Topor, M., Veggel, N., Yeung, S., Mellor, D., & Pfeiffer, N. Generalized Systematic Review Registration Form. MetaArXiv..

<https://doi.org/g5fj>

Statement of Deviations

1. Data extraction:

- Removed experimental manipulation and replaced with data collection method (survey vs experiment)
- Did not perform additional analyses on ‘study design type’ due to low agreement and being unnecessary for interpretation and ‘costs paid by whom?’ due to space limitation

2. Prompt GPT to include rationale on second round:

- This was removed, because adding this to the prompt changed GPT’s decisional output (low agreement with itself), but when the same prompt was used (without instructions to include rationale), GPT showed high internal agreement.

Preregistration

Databases: PubMed,
Scopus, Web of Science

Search dates: March-April
2025

Inclusion Criteria:

- Peer-reviewed journal articles, including empirical studies, theoretical papers, and reviews.
- Studies examining future thinking in relation to temporally distant outcomes.
- Research investigating how future thinking shapes decision-making, intentions, or policy support.
- Focus on long-term collective action contexts (e.g., climate change, resource management, infectious disease control, collective bargaining).

Exclusion Criteria:

- Studies focusing primarily on how threats shape delay discounting without linking it to collective action.
- Studies that focus on future thinking regarding individual outcomes (e.g., personal financial decisions, health choices) only.
- Philosophical papers that involve normative assertions of how people should value the future.
- Retracted articles.

Publication & Accessibility:

- Articles must be published in English.
- Conference abstracts, book chapters, and unpublished dissertations will be excluded.

Databases and Query Strings

Web of Science

TS=("future thinking" OR "distant future thinking" OR "future perceptions" OR "distant future perceptions" OR "time perception" OR "long-term thinking" OR "future discounting" OR "delay discounting" OR "foresight" OR "anticipation" OR "collective future thinking" OR "risk perception theory" OR "temporal discounting" OR "hyperbolic discounting" OR "linear discounting" OR "exponential discounting" OR "construal level theory" OR "temporal construal theory" OR "prospective cognition" OR "future consciousness" OR "future-oriented thinking" OR "future-self continuity" OR "psychological distance" OR "episodic future thinking" OR "temporal distance" OR "utopian thinking" OR "dystopian" OR "apocalyptic") AND TS=("decision making" OR "choice" OR "judgment" OR "judgement" OR "choice behaviour" OR "choice behavior" OR "risk perception" OR "threat perception" OR "risk assessment" OR "self-regulation" OR "self-control") AND TS=("collective action" OR "intergenerational collective action" OR "social dilemma" OR "public goods problem" OR "cooperation" OR "prosocial behaviour" OR "prosocial behavior" OR "prosociality" OR "sustainability" OR "climate change" OR "global warming" OR "environmental sustainability" OR "infectious disease" OR "pandemic response" OR "Covid-19" OR "public health intervention" OR "antimicrobial resistance" OR "antibiotic resistance" OR "drug-resistant infections" OR "vaccination" OR "herd immunity" OR "conservation" OR "political participation" OR "collective bargaining" OR "futurability")

Scopus

(TITLE-ABS-KEY("future thinking" OR "distant future thinking" OR "future perceptions" OR "distant future perceptions" OR "time perception" OR "long-term thinking" OR "future discounting" OR "delay discounting" OR "foresight" OR "anticipation" OR "collective future thinking" OR "risk perception theory" OR "temporal discounting" OR "hyperbolic discounting" OR "linear discounting" OR "exponential discounting" OR "construal level theory" OR "temporal construal theory" OR "prospective cognition" OR "future consciousness" OR "future-oriented thinking" OR "future-self continuity" OR "psychological distance" OR "episodic future thinking" OR "temporal distance" OR "utopian thinking" OR "dystopian" OR "apocalyptic")) AND (TITLE-ABS-KEY("decision making" OR "choice" OR "judgment" OR "judgement" OR "choice behaviour" OR "choice behavior" OR "risk perception" OR "threat perception" OR "risk assessment" OR "self-regulation" OR "self-control")) AND (TITLE-ABS-KEY("collective action" OR "intergenerational collective action" OR "social dilemma" OR "public goods problem" OR "cooperation" OR "prosocial behaviour" OR "prosocial behavior" OR "prosociality" OR "sustainability" OR "climate change" OR "global warming" OR "environmental sustainability" OR "infectious disease" OR "pandemic response" OR "Covid-19" OR "public health intervention" OR "antimicrobial resistance" OR "antibiotic resistance" OR "drug-resistant infections" OR "vaccination" OR "herd immunity" OR "conservation" OR "political participation" OR "collective bargaining" OR "futurability"))

PubMed

("future thinking"[tiab] OR "distant future thinking"[tiab] OR "future perceptions"[tiab] OR "time perception"[mh] OR "long-term thinking"[tiab] OR "future discounting"[tiab] OR "delay discounting"[mh] OR "foresight"[tiab] OR "anticipation"[tiab] OR "collective future thinking"[tiab] OR "risk perception theory"[tiab] OR "temporal discounting"[tiab] OR "hyperbolic discounting"[tiab] OR "linear discounting"[tiab] OR "exponential discounting"[tiab] OR "construal level theory"[tiab] OR "temporal construal theory"[tiab] OR "prospective cognition"[tiab] OR "future consciousness"[tiab] OR "future-oriented thinking"[tiab] OR "future-self continuity"[tiab] OR "psychological distance"[tiab] OR "episodic future thinking"[tiab] OR "temporal distance"[tiab] OR "utopian thinking"[tiab] OR dystopian[tiab] OR apocalyptic[tiab]) AND ("decision making"[mh] OR choice[tiab] OR judgment[tiab] OR judgement[tiab] OR "choice behaviour"[tiab] OR "choice behavior"[tiab] OR "risk perception"[tiab] OR "threat perception"[tiab] OR "risk assessment"[mh] OR "self-regulation"[tiab] OR "self-control"[mh]) AND ("collective action"[tiab] OR intergenerational[tiab] AND "collective action"[tiab] OR "social dilemma"[tiab] OR "public goods problem"[tiab] OR cooperation[tiab] OR prosocial[tiab] OR sustainability[tiab] OR "climate change"[mh] OR "global warming"[tiab] OR "environmental sustainability"[tiab] OR "infectious disease"[tiab] OR "pandemic response"[tiab] OR "Covid-19"[mh] OR "public health intervention"[tiab] OR "antimicrobial resistance"[tiab] OR "antibiotic resistance"[tiab] OR "drug-resistant infections"[tiab] OR "vaccination"[mh] OR "herd immunity"[tiab] OR conservation[tiab] OR "political participation"[tiab] OR "collective bargaining"[tiab] OR futurability[tiab])

Human-AI Screening and Validation

Prompting GPT: A prompt based on the inclusion/exclusion criteria was applied to all abstracts using GPT-4o in two independent runs, with one run requiring a rationale for each decision. Discrepancies between the human reviewer and GPT, or between the two GPT runs, were resolved by evaluating GPT’s rationale against the inclusion criteria and the abstract content. The study selection process was fully reported and visualized using a PRISMA-ScR flow diagram. Additionally, studies identified during the preliminary search were retained even if they did not appear in the final database search results.

Prompt (same for round 1 and 2)

I will provide the title and abstract from a scholarly article. Your task is to assess whether the article meets all of the following criteria: First, if the abstract concerns an empirical study, it must measure or manipulate cognitions, perceptions, or narratives about the future, such as future thinking, future orientation, anticipated outcomes, psychological temporal distance, or future discounting. If the abstract concerns, theoretical or philosophical approaches it must explicitly address these constructs. Please exclude papers focused on foresight methods, scenario planning, or anticipatory governance, and studies proposing specific policy/action plans. Secondly, it addresses societal challenges requiring cooperation. This may include real-world challenges, such as climate change, resource conservation, vaccination and herd immunity, and controlling the spread of infectious disease. It may also focus on conceptual analyses of coordination dilemmas, societal risks, or other issues requiring sustained collective cooperation over time. Third, it involves prosocial decision-making, even if motivated by self-interest, including behaviors, intentions, moral reasoning, or strategies that support the collective good—often involving trade-offs or personal costs, such as inconvenience or delayed gratification. Fourth, for empirical papers, the study directly measures or tests attitudes (e.g., support for climate policies), intentions (e.g., willingness to reduce energy use), or behaviors (e.g., self-reported or observed mitigation actions). For theoretical or philosophical papers, it must conceptually address all of these dimensions. If the abstract meets all four of these criteria, annotate it with a 1. If any of the criteria are not met, annotate it with a 0. Please include a short rationale for your response.’

Human-AI Screening and Validation

Agreement and reliability with human screener

Round 1 GPT-4

% agreement = 89.6

Cohen’s Kappa = .43

Gwet’s Ac1 = .9

Table S1

Confusion Matrix: round 1 vs human

	GPT-4: 0	GPT-4: 1
Human: 0	1049	53
Human: 1	76	61

Note. Precision = .54, Recall =.45, F1 = .49. The human screener is not the ground truth (see summary on next page)

Round 2 GPT-4

% agreement = 89.5

Cohen’s Kappa = .43

Gwet’s Ac1 = .9

Table S2

Confusion Matrix: round 2 vs human

	GPT-4: 0	GPT-4: 1
Human: 0	1048	54
Human: 1	76	61

Note. Precision = .53, Recall =.45, F1 = .48. The human screener is NOT the ground truth (see summary on next page)

Human-AI Screening and Validation

Agreement and reliability between two rounds of GPT-4

% agreement = 99.1

Cohen’s Kappa = .95

Gwet’s Ac1 = .99

Table S3

Confusion Matrix: round 1 vs round 2

	Round 2: 0	Round 2: 1
Round 1: 0	1119	6
Round 1: 1	5	109

Summary of Human-AI screening metrics:

Across two screening rounds, GPT-4 showed highly consistent agreement with the human screener (Round 1: 89.6%, Round 2: 89.5%), with moderate Cohen’s κ (.43) and high Gwet’s AC1 (.90) in both rounds, indicating stable and reproducible alignment. Classification metrics were similarly consistent across rounds (precision $\approx$.53–.54, recall $\approx$.45, F1 $\approx$.48–.49). Although recall appeared modest when benchmarked against the human screener, this comparison is limited by the absence of a definitive ground truth.

Final inclusion decisions reflect adjudicated outputs following human-AI disagreement resolution. Disagreement resolution showed that some GPT-identified inclusions initially rejected by the human were subsequently retained, suggesting that apparent false positives partly reflect productive disagreements. In a dual-screening context, these discrepancies are informative, as they prompt adjudication and help identify potentially missed studies. Overall, GPT-4 functions as a consistent and conservative second screener. See file: resolve disagreements screening.xlsx for human-AI disagreement resolution and justification.

GPT-4 demonstrated extremely high internal reliability across two screening rounds, with 99.1% agreement, Cohen’s κ = .95, and Gwet’s AC1 = .99. These values indicate near-perfect consistency, showing that GPT-4 produces highly stable and reproducible screening decisions with minimal variation across repeated runs.

Data Extraction

Study Characteristics

- 'Article Type',
- 'Study order within article',
- 'Preregistration',
- 'Country of Study',
- 'Sampling Methods'**,
- 'Sample Size',
- 'Participant Characteristics',
- 'Study Settings',
- 'Theoretical Frameworks',
- 'Type of Collective-Action Problem',
- 'Future Cognition Independent Variables',
- 'Other Measures',

- 'Study Design Type',
- 'Costs paid by whom?', (Self, collective, or both)*,
- 'Costs involve delay discounting?'*,
- 'Outcome measure benefits whom? (Self, collective, or both)*',
- Outcome of Results-supportive, non-supportive, mixed

AI-generated and validated

- 'Summary Results',
- 'Significant Predictors',
- 'Significant Mediators',
- 'Significant Moderators',
- 'Dependent Variable'

*Used to assist in reasoning during data extraction, no further analysis

**Limited analysis due to complexity (predominately convenience)

Validation

A second human rater was used to assess the reliability of both the study characteristic coding and outcome coding. For study characteristics, the rater independently coded a subset of 20 quantitative studies, achieving 84% agreement with the primary coder. Disagreements were mainly due to difficulties distinguishing between-subject vs. mixed designs and in classifying sampling methods; because of this ambiguity, sampling method classifications were excluded from the final analysis. (data files on Github)

For outcome coding, which classified each study–theory pairing as supportive, non-supportive, or mixed, two independent coders were involved. The second coder independently coded a subset of studies (20–38%), with intentional oversampling of studies reporting mixed or null findings to ensure robust evaluation of more ambiguous cases. Inter-coder agreement was 80%, and all discrepancies were resolved through discussion to reach consensus. (data files on Github)

AI Summaries of Results

- LLM platforms for generating results summaries: Microsoft Copilot and DeepSeek
- Prompt (2x): “Please summarise the results of this study in 6 sentences, including key quantitative outcomes.” (*note that there is no fixed length for “sentences”, and they could be longer or shorter depending on number of results to report*)

Process of generating summaries:

Study results sections were copied pasted directly into LLM chats, and summaries were generated in 2 rounds, producing 2 summaries for each article. I then applied GPT 4 through an R script to rate the similarity of the two summaries on a scale from 1-5 (2x) with GPT 4 showing high internal agreement and reliability with its ratings (% agreement = 93.8 and Cohen’s Kappa = .87). Eleven ratings were between 1-2 and were found to be due to mismatched (pasted in wrong row) or missing results summaries. In those cases, the AI was prompted to summarise the results again, and the LLM was prompted to provide a similarity score for the updated summaries.

GPT-4 was instructed to rate the similarity between two texts on a 5-point Likert scale (1 = highly dissimilar, 5 = highly similar), using the prompt: “Please rate the similarity of these two texts on a scale from 1–5 where 1 is highly dissimilar and 5 is highly similar. Please do not provide any information except the rating.” The 5-point scale was selected following preliminary comparisons with alternative scales (1–3 and 1–7), which indicated that it provided the most appropriate balance of granularity and reliability.

Human-AI extraction of predictors, mediators, moderators, and DVs (significant statistical paths):

Pasting the results summaries into LLM chats, it was prompted to extract all significant paths, “Please extract all significant paths including predictors, mediators, moderators, and dependent variables.” Only paths with intentions, behaviours, or policy as the DVs were extracted and analysed. This process was done in 2 rounds to ensure consistency.

AI summaries and significant statistical paths were randomly sampled and checked for accuracy and consistency against the source article at multiple stages during analysis.

Classifying Study Outcomes

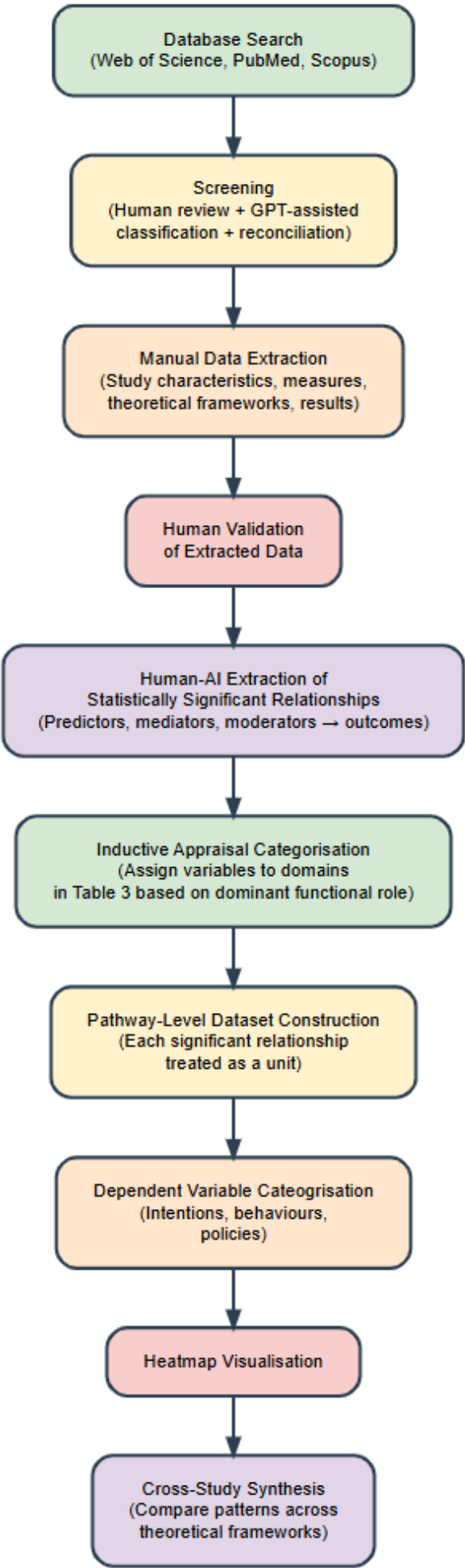
Outcomes were coded as supportive when the pattern of results aligned with theoretical predictions. Outcomes were coded as **non-supportive** when results did not conform to theoretical predictions, including null effects or effects in the wrong direction. **Mixed** outcomes were coded when support was partial or inconsistent, such as when predicted effects emerged for some dependent variables but not others, or when effects were observed in one sample or subgroup but not across all samples. Outcomes were coded as **supportive** when the results aligned with the theoretical predictions. Two independent coders were involved in outcome coding.

Inductive Categories of Appraisal Domains

Linking Appraisal Domains to Data Extraction

The inductive domains described in Table 3 and the Methods section of the main manuscript were applied during data extraction to categorise predictors, mediators, and moderators. Variables identified within each study were assigned to domains based on their dominant functional role within the reported statistical relationships (e.g., risk perception → Threat Knowledge; perceived personal impact → Personal Stake; perceived controllability → Efficacy). This approach enabled the integration of conceptually similar variables across different theoretical traditions within a shared appraisal framework.

Fig. S1
Workflow Chart



Main Results

Pages 15-61 are tables and heatmaps of significant predictors, mediators, and moderators for each theoretical framework. Heatmaps include DVs.

Due to the greater size of the literature on Construal Level Theory, this literature is further broken down by collective action domain (climate change and infectious disease control) and additionally by surveys vs experiments in the case of climate change.

Heatmap Construction

Heatmaps were constructed to visualise the co-occurrence of feature variables, including significant predictors, mediators, moderators, and dependent variables, across unique statistical pathways. The primary unit of analysis was the significant statistical pathway (e.g., predictor → DV, or mediated relationship), rather than the study. There could therefore be multiple pathways from a single study. Values are binarized to indicate the presence or absence of each feature within a given pathway.

To determine ordering, Jaccard distance (for binary data) was computed, and hierarchical clustering (average linkage) was applied where sufficient data were available. The resulting cluster ordering were used to arrange both row (features) and columns (pathways).

Row clusters were derived from hierarchical clustering solution, with the optimal number of clusters determined using silhouette analysis. Specifically, average silhouette widths were computed across a range of candidate cluster solutions, and the value of k that maximised silhouette width was selected. Cluster boundaries were then indicated by visual gaps in the heatmap. Additionally, sets of features with identical co-occurrence patterns across pathways (“bundles”) were identified and annotated, provided they co-occurred across at least two features.

Columns were annotated by data collection method (i.e., survey vs. experiment) except for Construal Level Theory as applied to climate change in which separate heatmaps were created. Heatmaps were rendered using a binary colour scheme (absent vs. present), with missing values shown in grey. These procedures were applied across all heatmaps.

Bundles capture repeated co-occurrence patterns among features by grouping rows that share identical binary profiles across pathways. Only groups comprising more than one feature and exhibiting at least two co-occurring presences were retained. Each bundle is annotated with its size (number of features) and support (number of pathways in which the share pattern is present), enabling identification of stable, recurring feature configurations.

Note: Clustering was used solely to improve visual organisation of variables within heatmaps and does not form the basis for theoretical or inferential claims. All substantive interpretations are based on the presence/absence and co-occurrence of variables across pathways, independent of cluster structure.

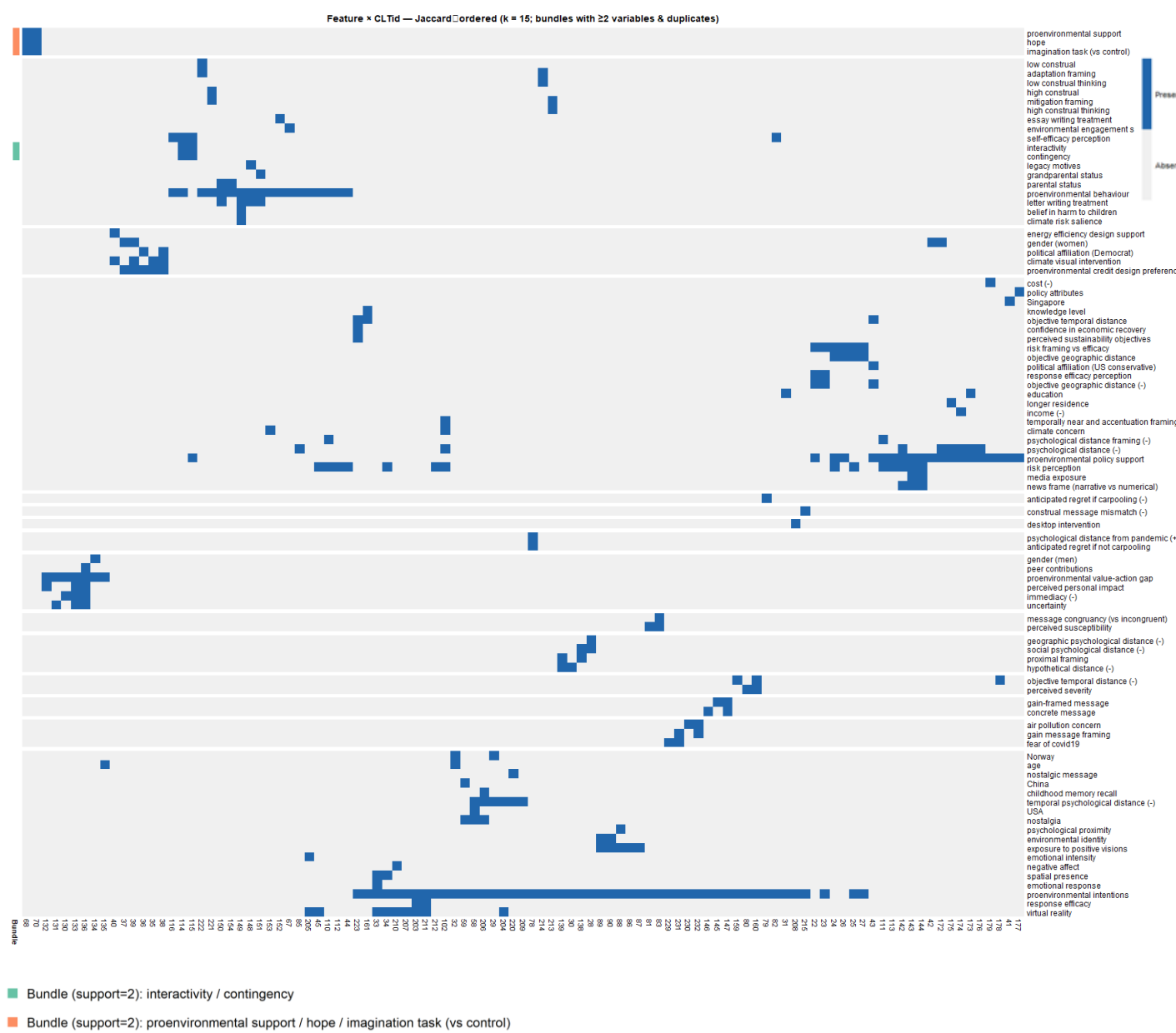
Table S4:
Construal Level Theory (climate change experiments only): significant predictors, mediators, and moderators by inductive category

Threat Knowledge	Personal Stake	Efficacy	Emotions	Future Cognition	Identity/ Reputation	Communicative Stimuli	Sociodemographic Factors
climate risk salience	belief in harm to children	contingency	climate concern	anticipated regret if carpooling (-)	environmental identity	adaptation framing	China
confidence in economic recovery	environmental engagement score		emotional intensity	anticipated regret if not carpooling	legacy motives	childhood memory recall	Norway
hypothetical distance (-)	geographic psychological distance (-)	peer contributions	emotional response	imagination task (vs control)	political affiliation (Democrat)	climate visual intervention	Singapore
knowledge level	grandparental status	policy attributes	hope	immediacy (-)	political affiliation (US conservative)	concrete message	USA
media exposure	longer residence	response efficacy	negative affect	objective temporal distance		construal message mismatch (-)	age
risk perception	objective geographic distance	response efficacy perception	nostalgia	objective temporal distance (-)		desktop intervention	education
uncertainty	objective geographic distance (-)	self-efficacy perception		proximal framing		essay writing treatment	gender (men)
high construal	parental status	cost (-)		temporal psychological distance (-)		exposure to positive visions	gender (women)
low construal	perceived personal impact					gain-framed message	income (-)
	perceived severity					letter writing treatment	
	perceived susceptibility					message congruancy (vs incongruent)	
	perceived sustainability objectives						
	psychological distance (-)					mitigation framing	
	psychological proximity					news frame (narrative vs numerical)	
	social psychological distance (-)					nostalgic message	
						risk framing vs efficacy	
	spatial presence					temporally near and accentuation framing	
	psychological distance framing (-)						
						virtual reality	

Note. Deduplicated variable representation across inductive categories (CLT climate change experiments). Displays all unique variables within significant pathways, and how they were categorised within this scheme.

Figure S2

Construal Level Theory (climate change experiments only): significant predictors, mediators, moderators, and dependent variables



Note. Heatmap CLT climate change experiments. Displays the co-occurrence of predictors, mediators, moderators, and dependent variables across extracted pathways. Blue indicates the presence of the variable and grey indicates absence of that variable. The far-left column represents bundles. Bundles are sets of variables that share identical co-occurrence patterns across paths.

Summary of table and heatmap

Table S4. CLT-based experimental studies show an emphasis on communicative stimuli, such as immersive formats like virtual reality, and personal stake, such as perceived susceptibility. There were relatively fewer measures of identity/reputation and emotions. Overall, the emphasis on framing and relevance reflects strategies to reduce psychological distance and increase urgency for climate action.

Figure S1. Across pathways, communicative stimuli variables frequently co-occur with temporal psychological distance, efficacy, and emotions in climate communication experiments. Immersive modalities (e.g., virtual reality) appear less frequently but tend to co-occur with emotional and distance-related variables. The heatmap indicates that the effects of messaging strategies and psychological distance are embedded within multi-component pathways.

Table S5

Construal Level Theory (climate change experiments): relabelling of dependent variables for heatmap

Relabel (dv.k2)	Original label (dv.k)
energy efficiency design support	energy efficiency design preference
proenvironmental behaviour	donation, donation amount, donation to environ organisation, low impact debrief email, mitigation behaviour, problem-solving coping, proenvironmental behaviour
proenvironmental credit design preference	climate credit design preference
proenvironmental intentions	behavioural intention, behavioural intentions, carpooling willingness, eco-friendly supply choice, eco-tech use intentions, individual/collective intentions, low-carbon behavioural intention, mitigation intentions, proenvironmental intentions, proenvrionmental intentions (plastic use), willingness to pay for biodiversity measures
proenvironmental policy support	policy preference, policy support
proenvironmental support	support for climate action
proenvironmental value-action gap	value-action gap

Note. Table of original dependent variables from experimental studies on climate change applying Construal Level Theory and relabels for consolidation in the heatmap.

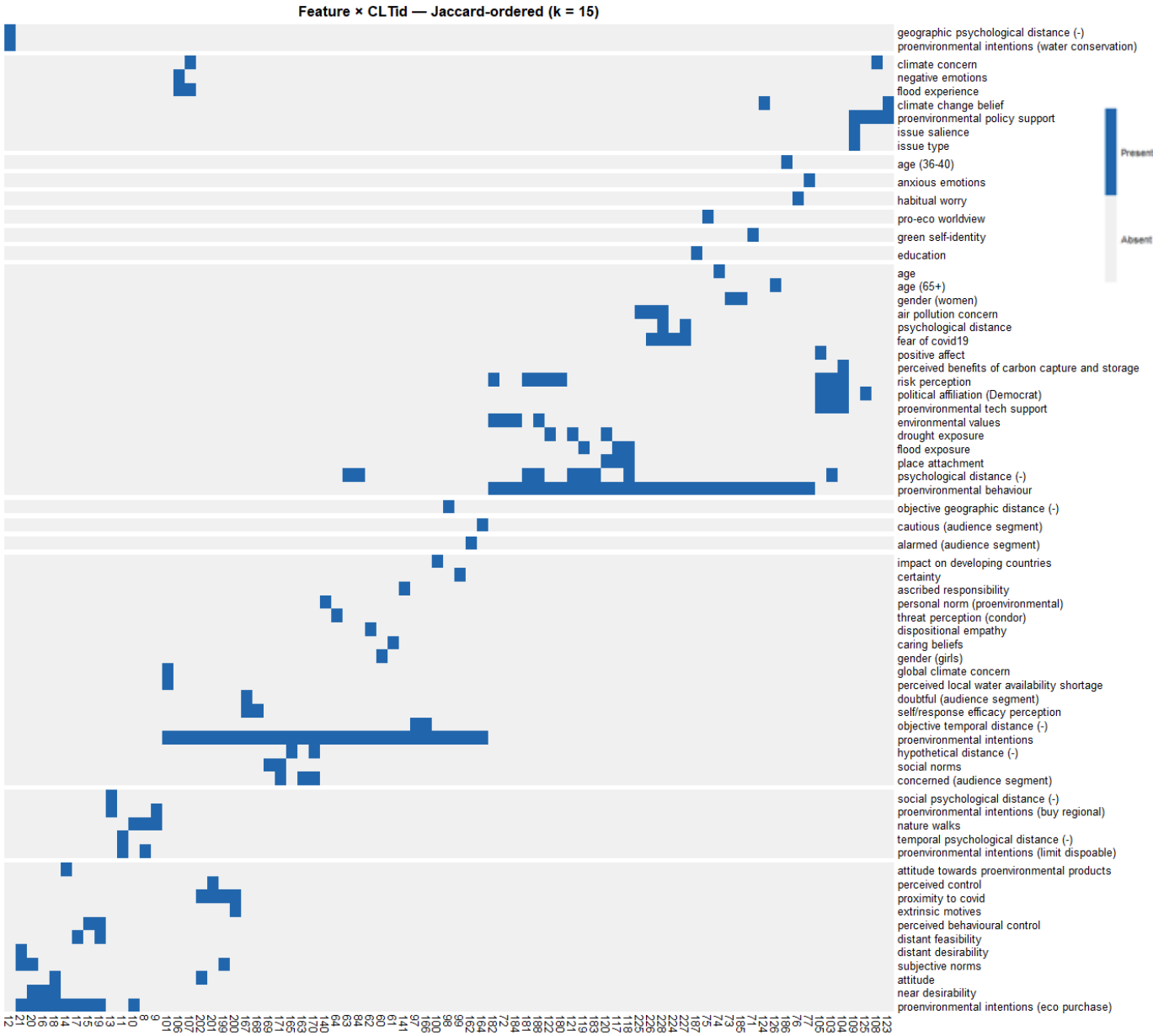
Table S6

Construal Level Theory (climate change surveys only): significant predictors, mediators, and moderators by inductive category

Personal Stake	Threat Knowledge	Efficacy	Emotions	Future Cognition	Identity/Reputation	Prosociality	Communicative Stimuli	Sociodemographic Factors
drought exposure	certainty	anticipated positive consequences	alarmed (audience segment)	objective temporal distance (-)	extrinsic motives	ascribed responsibility	outcome framing (gain)	age
flood experience	climate change belief	attitude towards proenvironmental products	anxious emotions	temporal psychological distance (-)	green self-identity	caring beliefs		age (36-40)
flood exposure	doubtful (audience segment)	distant desirability	attitude		political affiliation (democrat)	dispositional empathy		age (65+)
geographic psychological distance (-)	issue salience	distant feasibility	cautious (audience segment)			environmental values		education
impact on developing countries	issue type	near desirability	climate concern		subjective norms	personal norm (proenvironmental)		gender (girls)
nature walks	risk perception	perceived behavioural control	concerned (audience segment)			pro-eco worldview		gender (women)
objective geographic distance (-)	threat perception (condor)	perceived benefits of carbon capture and storage	fear of covid19					
place attachment	hypothetical distance (-)	perceived control	global climate concern					
proximity to covid		self/responsible efficacy perception	habitual worry					
psychological distance (-)			negative emotions					
social psychological distance (-)			positive affect					
perceived local water availability shortage								

Note. Deduplicated variable representation across inductive categories (CLT climate change surveys). Displays all unique variables within significant pathways, and how they were categorised within this scheme.

Figure S3
 Construal Level Theory (climate change surveys only): significant predictors, mediators, moderators, and dependent variables



Note. Heatmap CLT climate change surveys. Displays the co-occurrence of predictors, mediators, moderators, and dependent variables across extracted pathways. Blue indicates the presence of the variable and grey indicates absence of that variable. There are no bundles in this data.

Summary of table and heatmap

Table S6. Among CLT-based surveys, constructs pertaining to personal stake were the most common, and emotion measures were the second most common. Predictably, few studies used communicative stimuli, relative to CLT-based experiments. There was little overlap with other theoretical measures of future cognitions in this literature.

Figure S2. Across pathways, variables related to personal stake, emotions, identity/reputation, and threat knowledge are widely distributed and frequently co-occur. Stake constructs frequently align with identity markers such as green self-identity and environmental values, as well as social group affiliations, indicating that perceived personal impact is reinforced by both self-concept within a collective. Threat knowledge occasionally co-occur with efficacy beliefs, forming a risk–efficacy pairing that supports adaptive responses, though these links are less frequent. This demonstrates multi-component pathways that predict intentions, behaviours, or policy support.

Table S7
 Construal Level Theory (climate change surveys): relabelling of dependent variables for heatmap

Relabel (dv.k2)	Original label (dv.k)
proenvironmental behaviour	climate communication, environmental action, low-carbon behaviours, past environmental behaviour, proenvironmental behaviour
proenvironmental intentions	climate action intentions, mitigation intentions, proenvironmental intentions, proenvrionmental intentions, willingness to protect, willingness to reduce energy use, willingness to sacrifice
proenvironmental intentions (buy regional)	intention to buy regional products, regional product purchasing
proenvironmental intentions (eco purchase)	eco-label purchase intentions, proenvironmental puchase intentions
proenvironmental intentions (limit dispoable)	intention to limit disposable products
proenvironmental intentions (water conservation)	water conservation
proenvironmental policy support	policy support
proenvironmental tech support	support carbon capture and storage

Note. Table of original dependent variables from survey studies on climate change applying Construal Level Theory and relabels for consolidation in the heatmap.

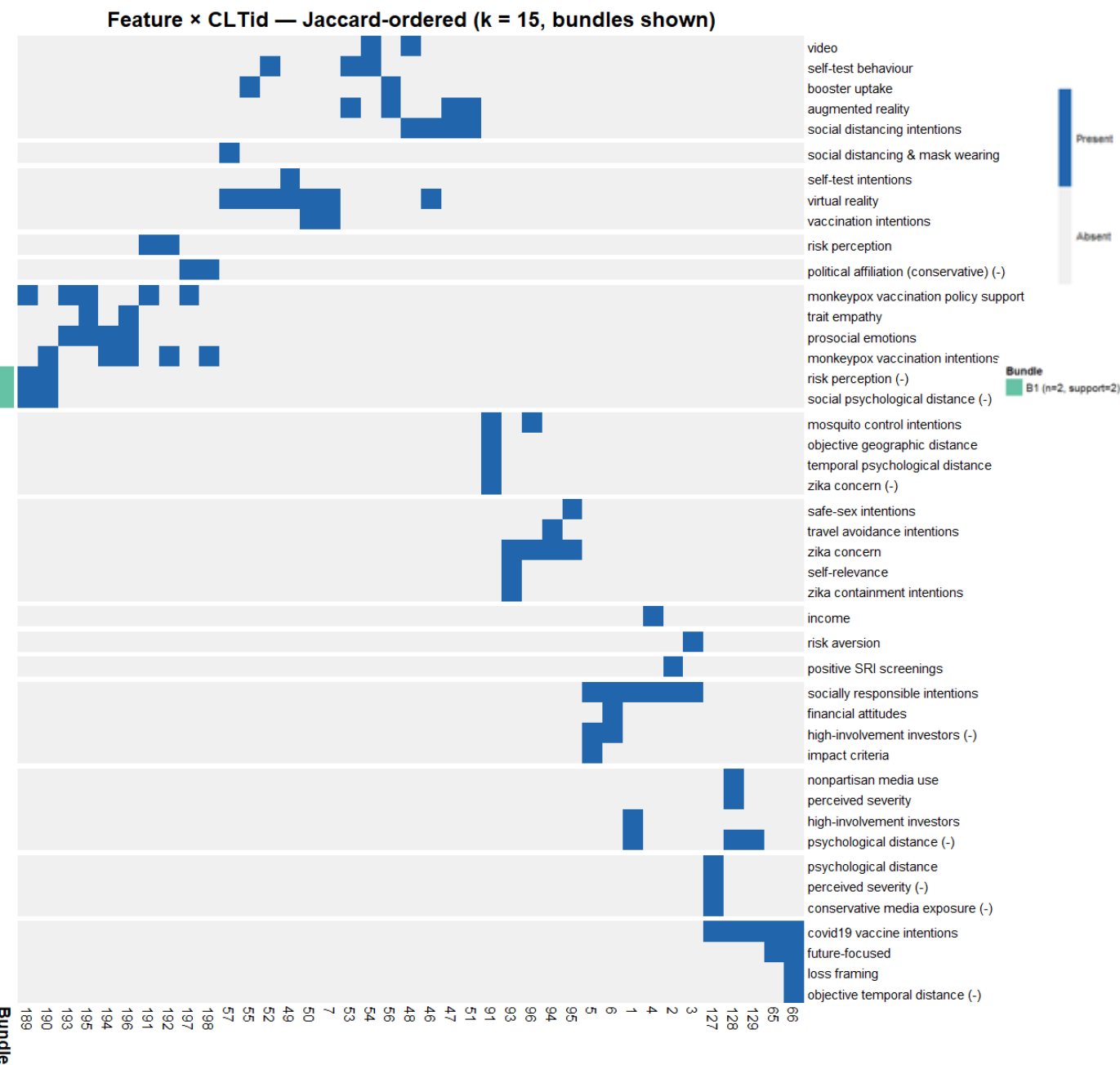
Table S8

Construal Level Theory (infectious disease control only): significant predictors, mediators, and moderators by inductive category

Threat Knowledge	Personal Stake	Efficacy	Emotions	Future Cognition	Identity/ Reputation	Communicative Stimuli	Prosociality	Sociodemographic Factors
certainty	high-involvement investors	controllability	financial attitudes	future-focused	political affiliation (conservative)	augmented reality	prosocial emotions	income
conservative media exposure (-)	high-involvement investors (-)	impact criteria	risk aversion	objective temporal distance (-)		gain message framing	trait empathy	
hypothetical distance (-)	objective geographic distance (-)	positive SRI screenings	zika concern	temporal psychological distance (-)		loss framing		
nonpartisan media use	psychological distance					loss message		
perceived severity	psychological distance (-)					video		
perceived severity (-)	self-relevance					virtual reality		
risk perception	social psychological distance (-)							
risk perception (-)								
uncertainty								

Note. Deduplicated variable representation across inductive categories (CLT infectious disease control). Displays all unique variables within significant pathways, and how they were categorised within this scheme.

Figure S4
Construal Level Theory (infectious disease only): significant predictors, mediators, moderators, and dependent variables



Note. Heatmap CLT infectious disease control. Displays the co-occurrence of predictors, mediators, moderators, and dependent variables across extracted pathways. Blue indicates the presence of the variable and grey indicates absence of that variable. The far left column represents bundles. Bundles are sets of variables that share identical co-occurrence patterns across paths.

Summary of table and heatmap

Table S8. In health-related CLT studies, communicative stimuli is the most common variable type, reflecting the prevalence of experimental designs. Variables pertaining to personal stake were also common, which is consistent with other CLT studies, as it is part of the model’s dimensions. There was little overlap with other measures of future cognition.

Figure S3. Variables related to personal stake, communicative stimuli, emotions, prosociality, and efficacy beliefs are frequently co-occurring, indicating integrated multi-component configurations. Stake and framing variables often appear together, while emotional and identity-related constructs align within the same pathways. Threat knowledge occasionally co-occurs with efficacy variables. Overall, behavioural intentions appear embedded within pathways linking perceived stake, threat knowledge, and efficacy beliefs, particularly as these relate to vaccination decisions.

Table S9

Construal Level Theory (infectious disease control): relabelling of dependent variables for heatmap

Relabel (dv.k2)	Original label (dv.k)
booster uptake	booster uptake (1 week later)
covid19 vaccine intentions	vaccine intentions
monkeypox vaccination intentions	vaccine intentions
monkeypox vaccination policy support	vaccine policy support
mosquito control intentions	mosquito control intentions
proenvironmental policy support	policy preference
safe-sex intentions	safe-sex intentions
self-test behaviour	self-test behaviour (1 week later)
self-test intentions	self-test intentions
social distancing & mask wearing	social distancing & mask wearing
social distancing intentions	social distancing intentions
socially responsible intentions	willingness to pay
travel avoidance intentions	travel avoidance intentions
vaccination intentions	vaccination intention, vaccination intention (1 week later)
zika containment intentions	zika containment intentions

Note. Table of original dependent variables from studies on infectious disease control applying Construal Level Theory and relabels for consolidation in the heatmap.

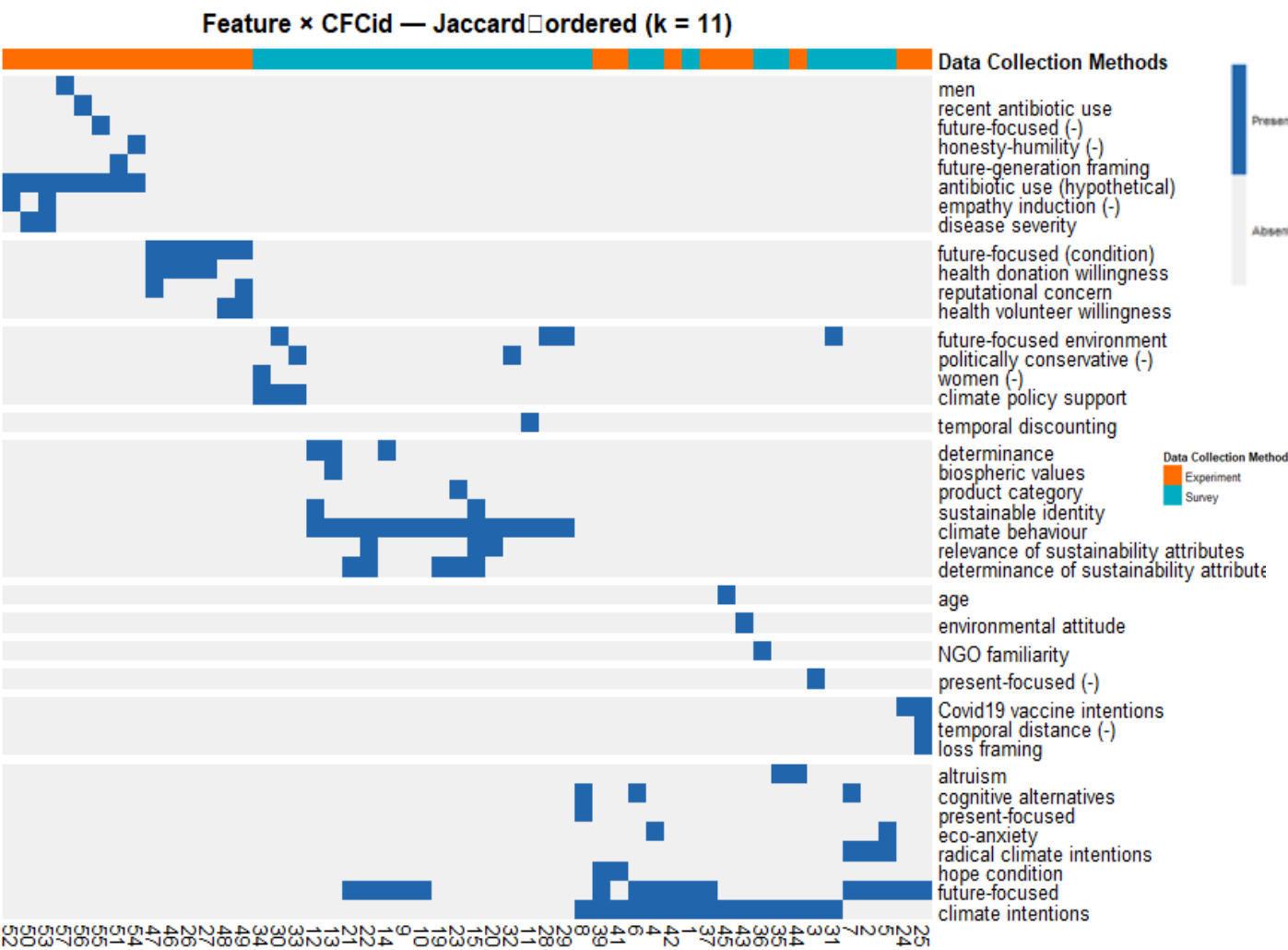
Table S10

Consideration of Future Consequences: significant predictors, mediators, and moderators by inductive category

Personal Stake	Emotions	Efficacy	Future Cognition	Identity/ Reputation	Prosociality	Communicative Stimuli	Sociodemographic Factors
determinance of sustainability attributes		NGO familiarity	cognitive alternatives	politically conservative (-)	altruism	temporal distance (-), loss framing	age
disease severity	environmental attitude		future-focused	reputational concern	empathy induction (-)	future-generation framing	men
disease severity (-)	hope condition		future-focused (-)	sustainable identity	honesty-humility (-)		women (-)
product category	humour condition		future-focused (condition)	biospheric values			
recent antibiotic use			future-focused environment				
relevance of sustainability attributes			present-focused				
			present-focused (-)				
			temporal discounting				

Note. Deduplicated variable representation across inductive categories (Consideration of Future Consequences). Displays all unique variables within significant pathways, and how they were categorised within this scheme.

Figure S5
 Consideration of Future Consequences: significant predictors, mediators, moderators, and dependent variables



Note. Heatmap Consideration of Future Consequences. Displays the co-occurrence of predictors, mediators, moderators, and dependent variables across extracted pathways. Blue indicates the presence of the variable and grey indicates absence of that variable. The top annotation row represents the data collection methods (orange = experiment; light blue = survey). There are no bundles in this data.

Summary of table and heatmap

Table S10. Studies testing Consideration of Future Consequences show an emphasis on personal stake, emotions, and identity/reputation. There are fewer constructs pertaining to efficacy.

Figure S4. The heatmap shows frequent co-occurrence between future cognitions, including future-focused orientations and conditions, and constructs tied to personal stake, identity/reputation, prosociality, and emotion. This shows that future focus operates in conjunction with appraisals of the self and others.

Table S11

Consideration of Future Consequences: relabelling of dependent variables for heatmap

Relabel (dv.k2)	Original label (dv.k)
Covid19 vaccine intentions	vaccine intentions
antibiotic use (hypothetical)	antibiotic use
climate behaviour	electricity conservation, environmental activism, environmental behaviour, environmental discussions, sustainable purchase
climate intentions	charitable donation, public and private intentions
climate policy support	climate policy support
health donation willingness	charitable donation, donation amount, gift giving (dictator game)
health volunteer willingness	volunteering
radical climate intentions	radical intentions

Note. Table of original dependent variables from studies applying Consideration of Future Consequences and relabels for consolidation in the heatmap.

Table S12

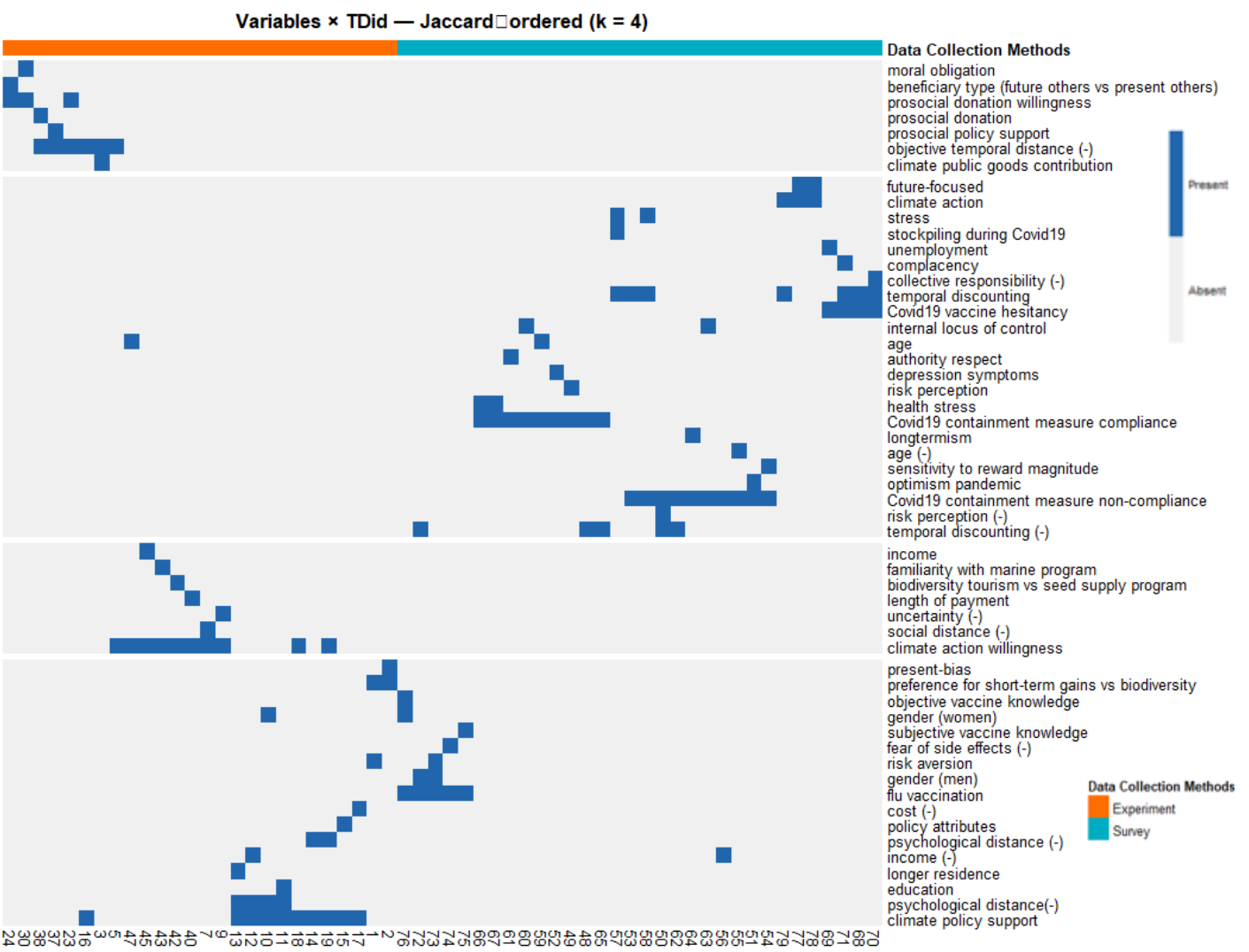
Temporal Discounting: significant predictors, mediators, moderators, and dependent variables by inductive category

Threat Knowledge	Personal Stake	Efficacy	Emotions	Future Cognition	Identity/ Reputation	Prosociality	Sociodemographic Factors
risk perception	beneficiary type (future others vs present others)	biodiversity tourism vs seed supply program	depression symptoms	future-focused	authority respect	collective responsibility (-)	age
	longer residence	climate policy support	fear of side effects (-)			moral obligation	age (-)
uncertainty (-)	psychological distance (-)	complacency	health stress	objective temporal distance (-)			education
	psychological distance(-)			present-bias			gender (men)
	sensitivity to reward magnitude	familiarity with marine program	risk aversion				gender (women)
	social distance (-)	internal locus of control	stress	temporal discounting			income
		length of payment		temporal discounting (-)			income (-)
		objective vaccine knowledge					unemployment
		policy attributes					
		subjective vaccine knowledge					

Note. Deduplicated variable representation across inductive categories (Temporal Discounting). Displays all unique variables within significant pathways, and how they were categorised within this scheme.

Figure S6

Temporal Discounting: significant predictors, mediators, moderators, and dependent variables



Note. Heatmap Temporal Discounting. Displays the co-occurrence of predictors, mediators, moderators, and dependent variables across extracted pathways. Blue indicates the presence of the variable and grey indicates absence of that variable. The top annotation row represents the data collection methods (orange = experiment; light blue = survey). There are no bundles in this data.

Summary of table and heatmap

Table S12. There is a greater representation of additional theoretical frameworks of future cognitions tested alongside Temporal Discounting in this literature. There is also a greater share of sociodemographic factors—notably age, education, and income—within the significant pathways. Personal stake, efficacy, and emotions are also prominent constructs. Identity/reputation and prosocial constructs were less common among the significant pathways.

Figure S5. The heatmap indicates that positive engagement with long-term collective-action emerges from an integrated interplay between factors. Future-oriented thinking and efficacy beliefs jointly support prosocial choices, with attention to long-term outcomes aligning with perceived controllability. Emotions, such as optimism and stress, interact with identity and social connectedness to shape motivations, although stress can produce byproduct effects that can either promote or hinder long-term collective action. Perceived stakes, threat knowledge, and efficacy knowledge are also embedded in pathways that link future cognitions to long-term engagement in long-term collective action.

Table S13

Temporal Discounting: relabelling of dependent variables for heatmap

Relabel (dv.k2)	Original label (dv.k)
Covid19 containment measure compliance	containment measure compliance, greater social distancing, lockdown compliance, mask use
Covid19 containment measure non-compliance	containment measure non-compliance, intentions for outdoor activities, lower adherence to social distancing, lower physical distancing
Covid19 vaccine hesitancy	vaccine hesitancy
climate action	electricity conservation, environmental behaviour
climate action willingness	annual willingness to pay, willingness to act, willingness to pay
climate policy support	policy preference, policy support
climate public goods contribution	cooperative investment
flu vaccination	vaccination
none	concern, lower temporal discounting, moral concern, moral obligation
preference for short-term gains vs biodiversity	preference for fast growing trees
prosocial donation	charitable donation pledge
prosocial donation willingness	monetary allocation, prosocial generosity
prosocial policy support	policy attitude
stockpiling during Covid19	stockpiling

Note. Table of original dependent variables from studies applying Temporal Discounting and relabels for consolidation in the heatmap.

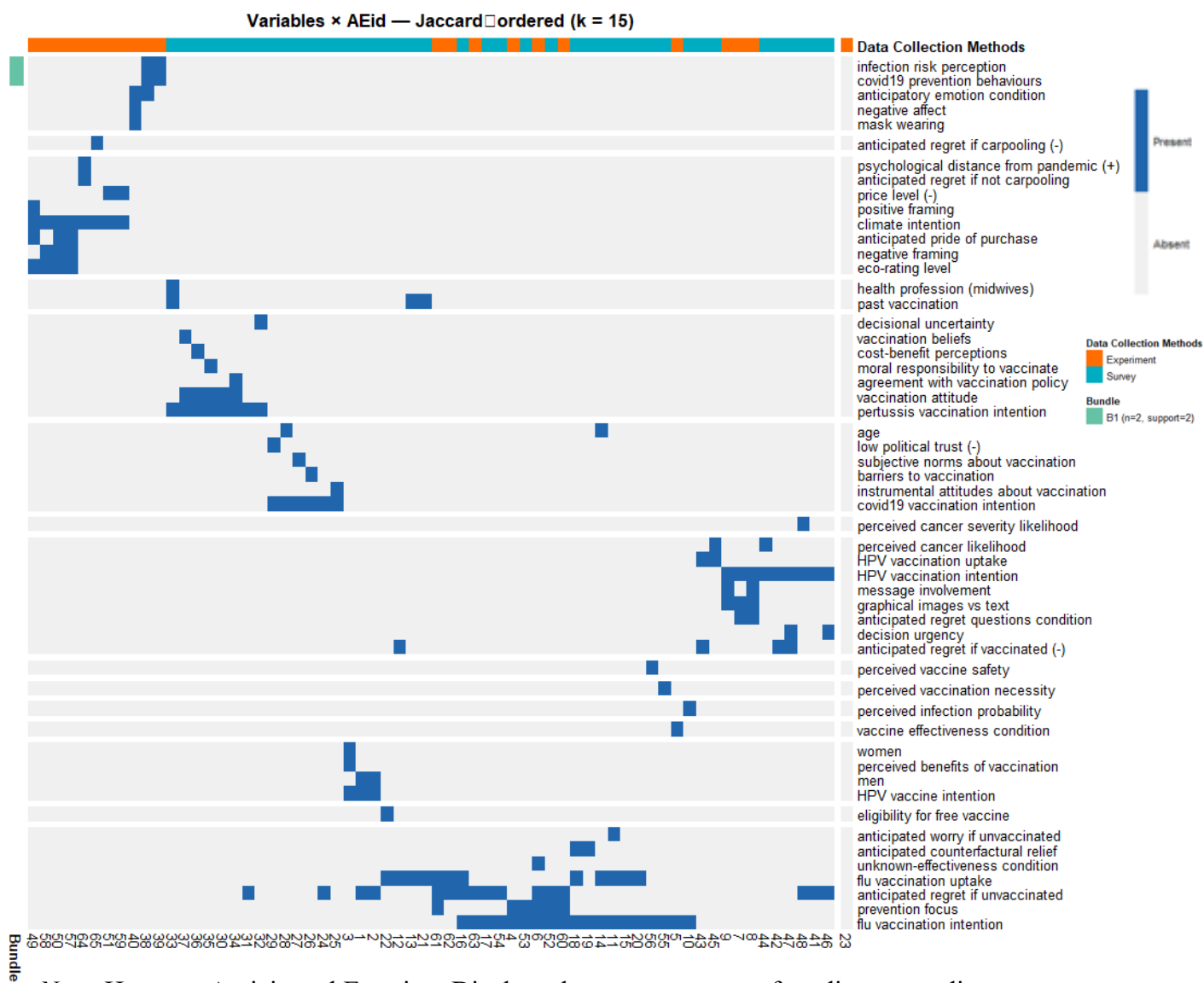
Table S14

Anticipated Emotion: significant predictors, mediators, moderators, and dependent variables by inductive category

Threat Knowledge	Personal Stake	Efficacy	Emotions	Future Cognitions	Identity/ Reputation	Prosociality	Sociodemographic Factors	Communicative Stimuli
decisional uncertainty	infection risk perception	agreement with vaccination policy	negative affect	anticipated counterfactual relief	low political trust (-)	moral responsibility to vaccinate	age	decision urgency
	message involvement	barriers to vaccination	prevention focus	anticipated pride of purchase	subjective norms about vaccination		health profession (midwives)	eco-rating level
	past vaccination	cost-benefit perceptions		anticipated regret if carpooling (-)			men	graphical images vs text
	perceived cancer likelihood	eligibility for free vaccine		anticipated regret if not carpooling			women	negative framing
	perceived cancer severity likelihood	flu vaccination intention		anticipated regret if unvaccinated				positive framing
	perceived infection probability	instrumental attitudes about vaccination		anticipated regret if vaccinated (-)				
		perceived benefits of vaccination		anticipated regret questions condition				
		perceived vaccination necessity		anticipated worry if unvaccinated				
		perceived vaccine safety		anticipatory emotion condition				
		price level (-)		psychological distance from pandemic (+)				
		unknown-effectiveness condition						
		vaccination attitude						
		vaccination beliefs						
		vaccine effectiveness condition						

Note. Deduplicated variable representation across inductive categories (Anticipated Emotion). Displays all unique variables within significant pathways, and how they were categorised within this scheme.

Figure S7
 Anticipated Emotion: significant predictors, mediators, moderators, and dependent variables



Note. Heatmap Anticipated Emotion. Displays the co-occurrence of predictors, mediators, moderators, and dependent variables across extracted pathways. Blue indicates the presence of the variable and grey indicates absence of that variable. The top annotation row represents the data collection methods (orange = experiment; light blue = survey). There are no bundles in this data. Note: For ID 23, there were no main effects predicting intentions, behaviour, or policy support (Okuno et al., 2021).

Summary of table and heatmap

Table S14. This literature focuses on decisions about vaccination. As such, there is an emphasis on anticipated vaccine outcomes based on vaccine attitudes and beliefs (efficacy) and its association with anticipated regret, anticipated worry, and counterfactual belief. There are also studies focused on adherence to COVID-19 social distancing guidelines.

Figure S5. Anticipated regret plays a key role by linking perceptions of risk and vaccine efficacy to stronger intentions for vaccination uptake. Communicative stimuli (e.g., gain/loss framing, fear appeals, and visual cues) was also associated with emotional forecasting and intentions. There were different pathways associated with intentions between men versus women, and older age was associated with stronger intentions to vaccinate. Factors that reduce barriers, such as eligibility for free vaccine, were also with intentions. Low political trust were associated with reduced intentions for COVID-19 vaccination.

Table S15

Anticipated Emotion: relabelling of dependent variables for heatmap

Relabel (dv.k2)	Original label (dv.k)
HPV vaccination intention	HPV vaccination intention
HPV vaccination uptake	HPV vaccination uptake
HPV vaccine intention	HPV vaccine intention (cancer), HPV vaccine intention (genital warts)
climate intention	carpooling willingness, sustainable purchase intention
covid19 prevention behaviours	covid prevention behaviours
covid19 vaccination intention	covid vaccination intention, provaccine group
flu vaccination intention	flu vaccination decision, flu vaccination intention, flu vaccination willingness
flu vaccination uptake	flu vaccination uptake
mask wearing	mask wearing
none	none
pertussis vaccination intention	pertussis vaccination intention

Note. Table of original dependent variables from studies applying Anticipated Emotion and relabels for consolidation in the heatmap.

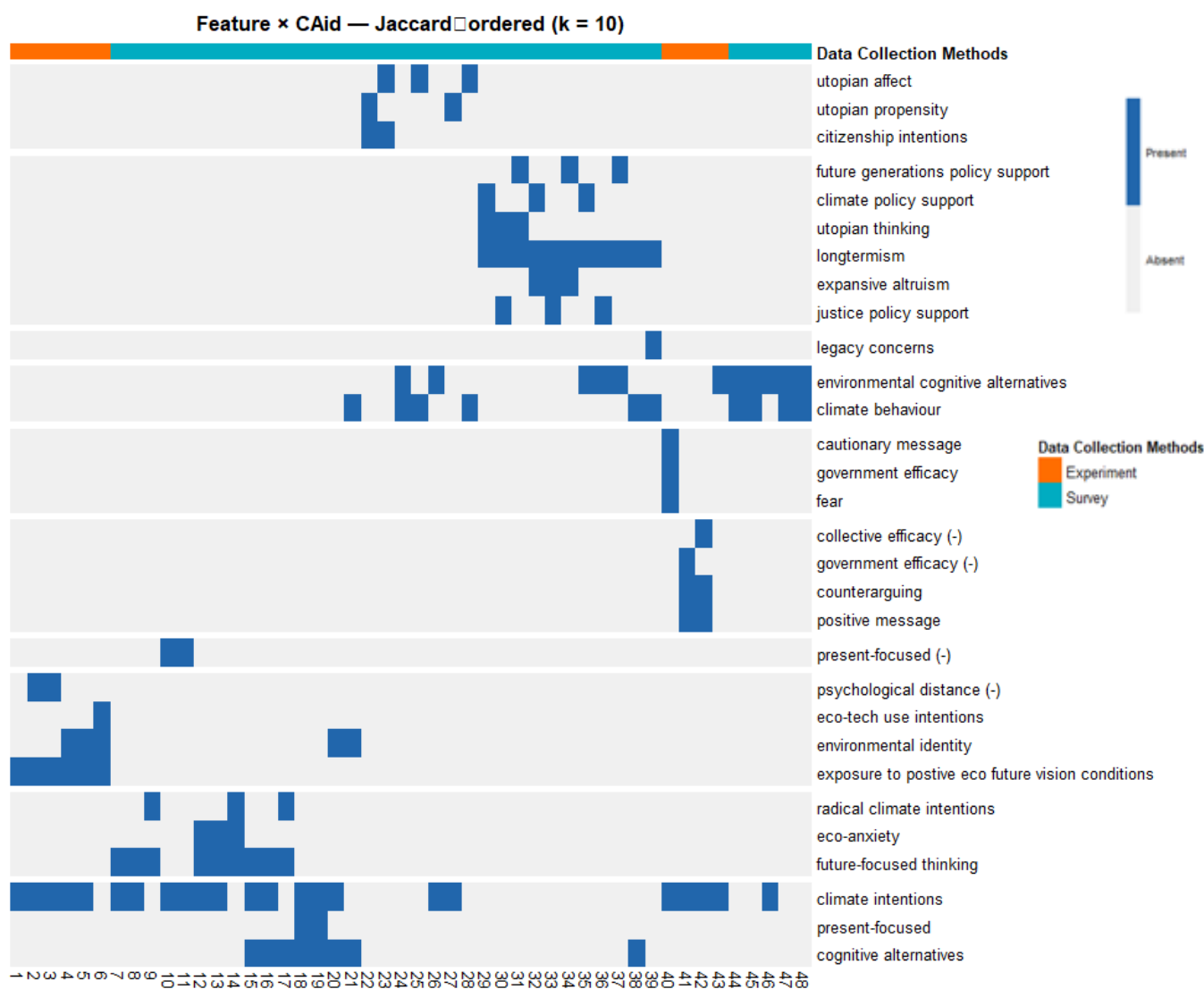
Table S16

Cognitive Alternatives: significant predictors, mediators, moderators, and dependent variables by inductive category

Threat Knowledge	Personal Stake	Efficacy	Emotions	Future Cognitions	Identity/ Reputation	Pro sociality	Communicative Stimuli
counterarguing	psychological distance (-)	collective efficacy (-)	eco-anxiety	cognitive alternatives	environmental identity	expansive altruism	cautionary message
		government efficacy	fear	environmental cognitive alternatives	legacy concerns		positive message
		government efficacy (-)	utopian affect	future-focused thinking			
				longtermism			
				present-focused			
				present-focused (-)			
				utopian propensity			
				utopian thinking			

Note. Deduplicated variable representation across inductive categories (Cognitive Alternatives). Displays all unique variables within significant pathways, and how they were categorised within this scheme.

Figure S8
Cognitive Alternatives: significant predictors, mediators, moderators, and dependent variables



Note. Heatmap Cognitive Alternatives. Displays the co-occurrence of predictors, mediators, moderators, and dependent variables across extracted pathways. Blue indicates the presence of the variable and grey indicates absence of that variable. The top annotation row represents the data collection methods (orange = experiment; light blue = survey). There are no bundles in this data.

Summary of table and heatmap

Table S16. Future cognition constructs predominate in the Cognitive Alternatives literature with considerable inclusion of constructs from Collective Futures Frameworks (e.g., longtermism). Communicative stimuli, identity/reputation, and emotions are also prominent. There was less attention towards threat knowledge, personal stake, efficacy, and prosociality.

Figure S6. Future-focused thinking constructs were positively associated with an interplay of variables to long-term collective action engagement, including identity/reputation, emotions, threat knowledge, and efficacy. Across studies, findings generally support a mechanistic role for Cognitive Alternatives in contrasting the status quo with a more desirable future.

Table S17

Cognitive Alternatives: relabelling of dependent variables for heatmap

Relabel (dv.k2)	Original label (dv.k)
citizenship intentions	citizenship intentions
climate behaviour	climate activism, consumer behaviour, individual proenvironmental actions, problem-focused coping, sign letter
climate intentions	climate action intentions, collective intentions, collective proenvironmental intentions, individual intentions, private intentions, public intentions, willingness for climate activism
climate policy support	climate policy support
eco-tech use intentions	eco-tech use intentions
future generations policy support	future generations policy support
justice policy support	justice policy support
radical climate intentions	radical intentions

Note. Table of original dependent variables from studies applying Cognitive Alternatives and relabels for consolidation in the heatmap.

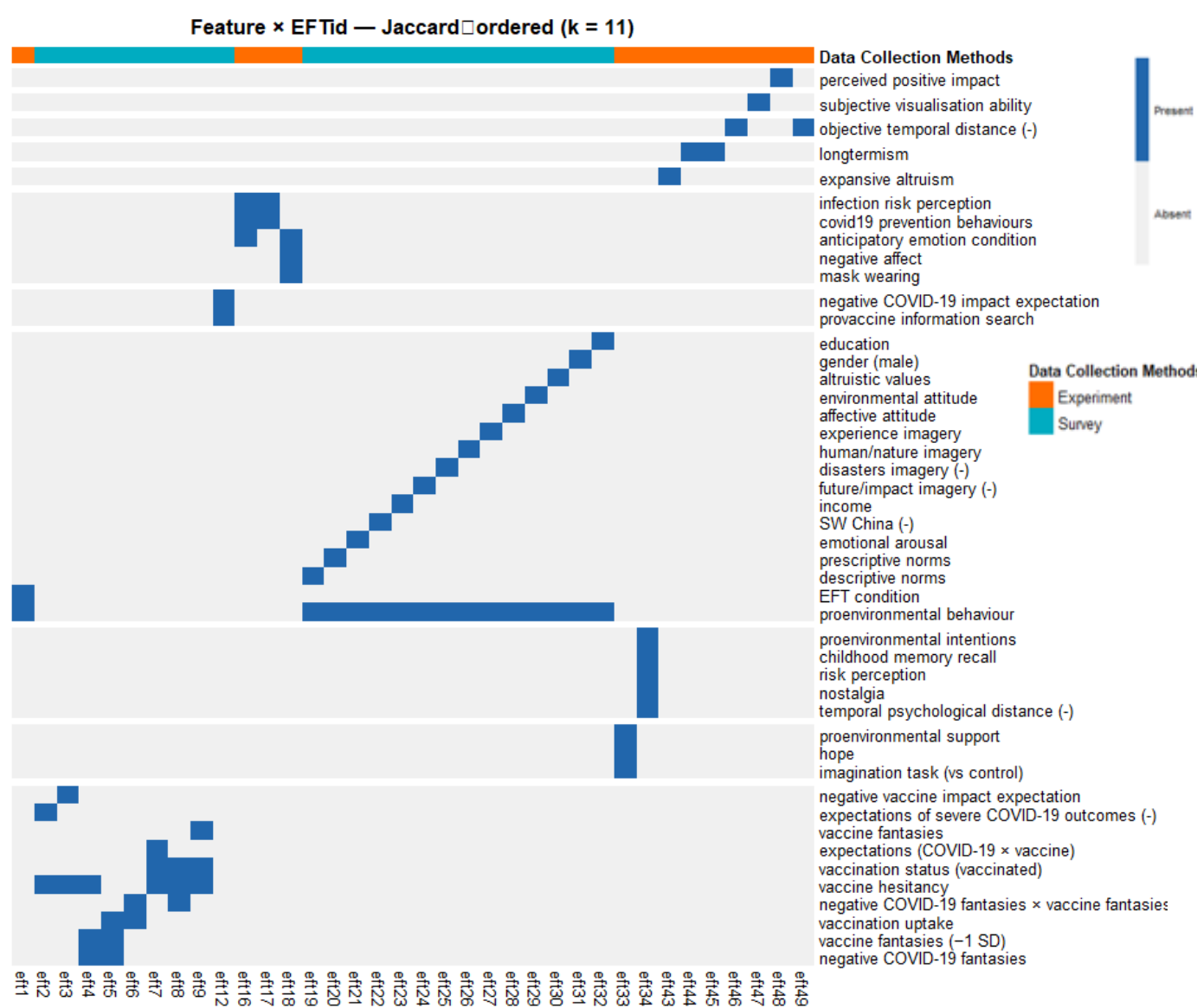
Table S18

Episodic Future Thinking: significant predictors, mediators, moderators, and dependent variables by inductive category

Threat Knowledge	Personal Stake	Efficacy	Emotions	Future Cognition	Identity/ Reputation	Pro sociality	Communicative Stimuli	Sociodemographic
disasters imagery (-)	negative COVID-19 impact expectation	expectations (COVID-19 × vaccine)	affective attitude	EFT condition	descriptive norms	altruistic values	childhood memory recall	China
infection risk perception	experience imagery	negative vaccine impact expectation	emotional arousal	anticipatory emotion condition	prescriptive norms	moral obligation	nostalgic message	SW China (-)
risk perception	human/nature imagery		environmental attitude	future/impact imagery (-)				USA
	infection risk perception		hope	imagination task (vs control)				education
	negative COVID-19 impact expectation (-)		negative affect	negative COVID-19 fantasies				gender (male)
	vaccination status (vaccinated)		nostalgia	negative COVID-19 fantasies × vaccine fantasies				income
	vaccination status (unvaccinated)			temporal psychological distance (-)				
				vaccine fantasies				
				vaccine fantasies (-1 SD)				

Note. Deduplicated variable representation across inductive categories (Episodic Future Thinking). Displays all unique variables within significant pathways, and how they were categorised within this scheme.

Figure S9
Episodic Future Thinking: significant predictors, mediators, moderators, and dependent variables



Note. Heatmap Episodic Future Thinking. Displays the co-occurrence of predictors, mediators, moderators, and dependent variables across extracted pathways. Blue indicates the presence of the variable and grey indicates absence of that variable. The top annotation row represents the data collection methods (orange = experiment; light blue = survey). There are no bundles in this data.

Summary of table and heatmap

Table S18. Future cognition was the most prominent construct, reflecting Episodic Future Thinking’s emphasis on vivid mental simulation of future outcomes through imagination. Efficacy, communicative stimuli, and emotions were also prominent among the significant pathways. In contrast, situational barriers identity/reputation, prosociality, and threat knowledge are less represented.

Figure S7. The heatmap shows pathways linking future cognitions, sometimes induced by communicative stimuli, with emotional factors (e.g., hope), highlighting their joint role in shaping intentions and behaviours.

Table S19

Episodic Future Thinking: relabelling of dependent variables for heatmap

Relabel (dv.k2)	Original label (dv.k)
covid19 prevention behaviours	covid prevention behaviours
mask wearing	mask wearing
none	moral concern, moral obligation
proenvironmental behaviour	donation, proenvironmental behaviour
proenvironmental intention	behavioural intention
proenvironmental intentions	mitigation intentions, proenvironmental intentions
proenvironmental support	support for climate action
prosocial donation willingness	prosocial generosity
provaccine information search	provaccine information search
vaccination uptake	vaccination uptake
vaccine hesitancy	vaccine hesitancy

Note. Table of original dependent variables from studies applying Episodic Future Thinking and relabels for consolidation in the heatmap.

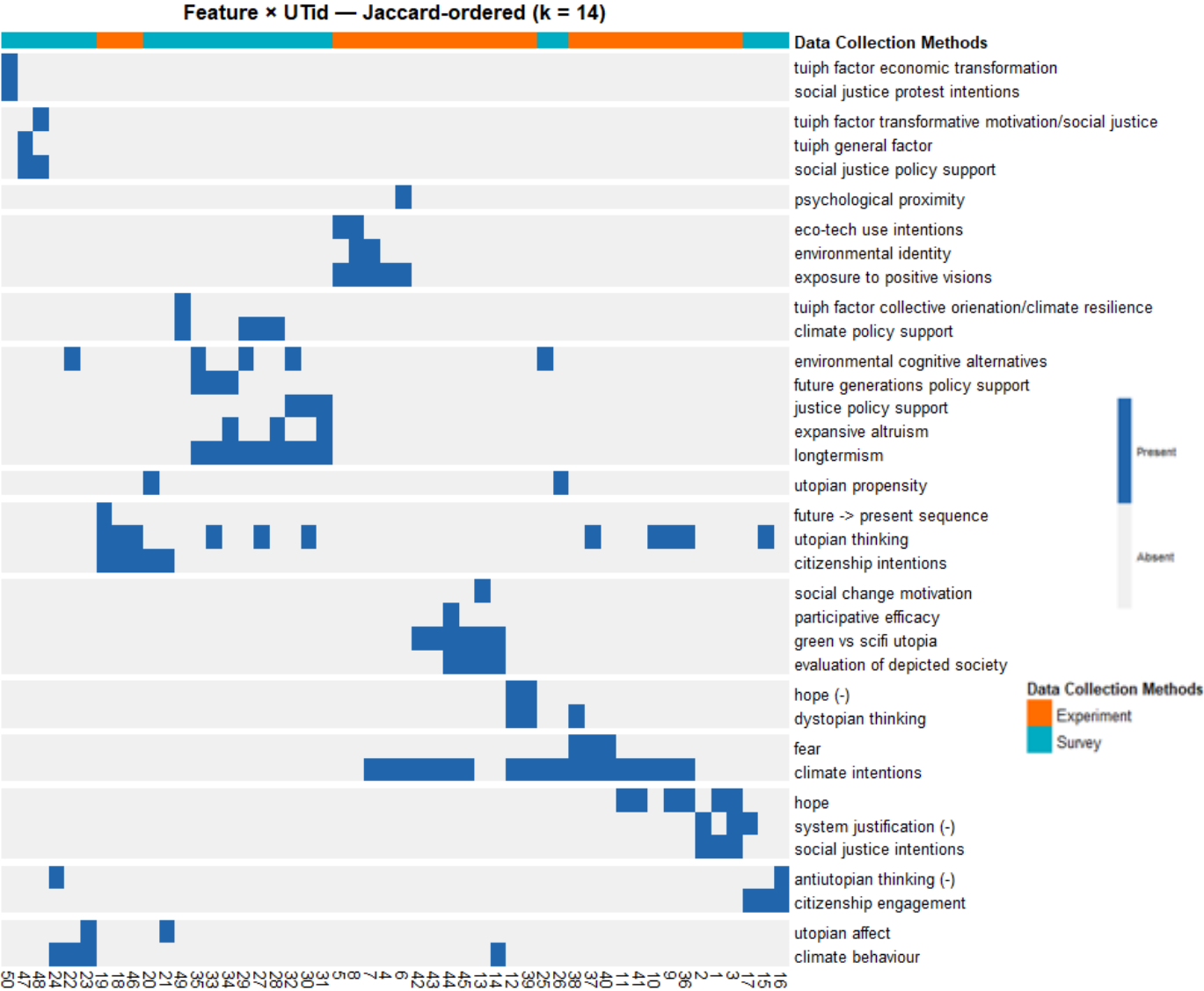
Table S20

Utopian Thinking: significant predictors, mediators, moderators, and dependent variables by inductive category

Threat Knowledge	Personal Stake	Efficacy	Emotions	Future Cognition	Identity/ Reputation	Prosociality	Communicative Stimuli
system justification (-)	psychological distance (-)	evaluation of depicted society	fear	antiutopian thinking (-)	environmental identity	expansive altruism	exposure to positive visions
		participative efficacy	hope	dystopian thinking		tuiph factor transformative motivation/social justice*	future -> present sequence
			hope (-)	environmental cognitive alternatives		tuiph factor* collective orienation/climate resilience	green vs scifi utopia
			tuiph affect*	longtermism			
				tuiph factor economic transformation *			
				tuiph general factor*			
				tuiph propensity*			
				utopian thinking			

Note. Deduplicated variable representation across inductive categories (Utopian Thinking). Displays all unique variables within significant pathways, and how they were categorised within this scheme. TUIPH stands for Transformative Utopian Impulse for Planetary Health and measures individual propensity to think about and engage in addressing social issues pertaining to climate change and human flourishing.

Figure S10
 Utopian Thinking: significant predictors, mediators, moderators, and dependent variables



Note. Heatmap Episodic Future Thinking. Displays the co-occurrence of predictors, mediators, moderators, and dependent variables across extracted pathways. Blue indicates the presence of the variable and grey indicates absence of that variable. The top annotation row represents the data collection methods (orange = experiment; light blue = survey). There are no bundles in this data.

Summary of table and heatmap

Table S20. Future cognition is a prominent construct in the Utopian Thinking literature. Emotions and communicative stimuli were also key variables in the experimental studies. Efficacy was moderately represented.

Figure S8. The heatmap reveals that future cognition constructs align with prosocial motivations and identity/reputation, while emotions such as hope and fear are reinforced through communicative stimuli that make future visions salient. Efficacy and personal stake are also represented in significant pathways.

Table S21

Utopian Thinking: relabelling of dependent variables for heatmap

Relabel (dv.k2)	Original label (dv.k)
citizenship engagement	citizenship engagement
citizenship intentions	citizenship intentions
climate behaviour	climate donation behaviour, individual environmental actions, individual environmental intentions
climate intentions	collective action intentions, collective environmental intentions, environmental citizenship, environmental striving, individual/collective intentions
climate policy support	climate policy support
eco-tech use intentions	eco-tech use intentions
future generations policy support	future generations policy support
justice policy support	justice policy support
social change motivation	social change motivation
social justice intentions	collective action intentions
social justice policy support	BLM and BBB policy support
social justice protest intentions	protest intentions

Note. Table of original dependent variables from studies applying UtopianThinking and relabels for consolidation in the heatmap.

Table S22

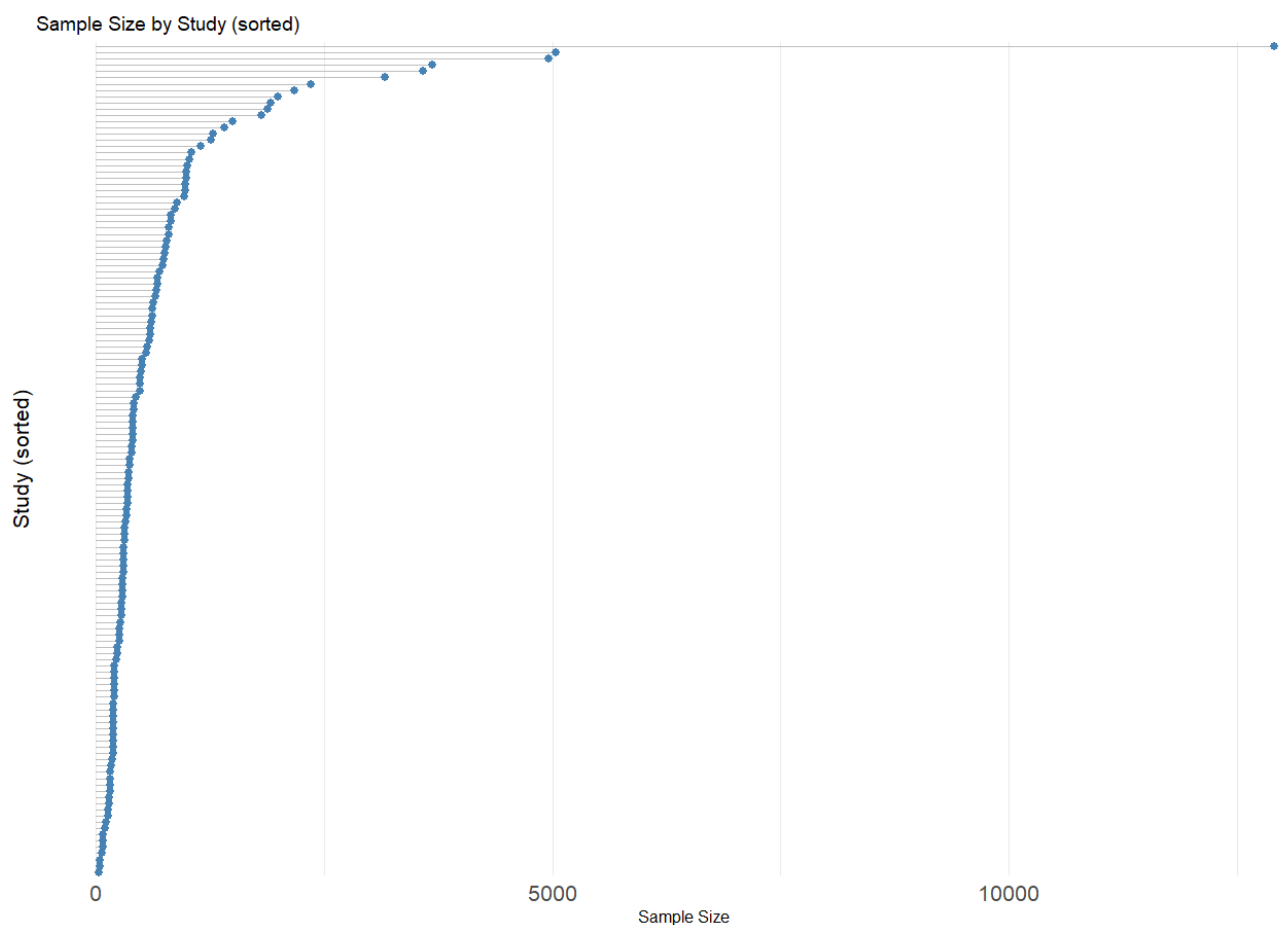
Four Leverage Points within the FOCAL Framework

Leverage Point	Core Mechanism	Key Conditions	Applications
1. Construal-coherent message framing	Messages are most effective when they maintain a single, internally coherent level of construal (abstract or concrete); mixing “why” and “how” cues disrupts motivational pathways and weakens effects.	Requires clear, consistent construal cues and sufficient perceived agency and relevance; coherence cannot compensate for low control or responsibility.	Use concrete “how-to” framing for actionable, near-term behaviours; abstract “why” framing for value formation and long-term goals; avoid mixed construal designs.
2. Reduce immediate burdens that prioritise the short term	Lower near-term costs, pressures, and stress that narrow temporal focus and suppress long-term action.	Most impactful in resource- or stress-constrained contexts; effective when short-term demands eclipse long-term intentions.	Reducing financial, logistical, or cognitive burdens; making benefits more immediate or visible; stabilising essential resources; simplifying guidance. May be combined with other leverage points to communicate policy changes and guidelines.
3. Activate future-focused and prosocial motivation	Engage long-term future orientation and concern for future others to increase the weight placed on distant outcomes.	Strongest among individuals already future-focused; enhanced when cues align with values, identity, and perceived collective responsibility.	Proximal-loss/long-term-gain framing; highlighting intergenerational consequences; emphasising community and collective welfare over time.
4. Foster constructive, meaningful, and feasible future simulations	Future simulations motivate behaviour through interacting long-term and near-term emotional pathways—hope and fear/anxiety (long-term anticipatory emotions) and anticipated pride/regret (near-term self-evaluative emotions)—supported by Cognitive Alternatives, which make futures feel realistically attainable or preventable.	Works when simulations enhance efficacy and feasibility; when threats feel preventable and desired futures feel attainable; when simulations align with identity and values; when hope does not suppress functional fear; when near-term anticipated emotions have clear behavioural scripts; and when Cognitive Alternatives support imagining realistic departures from the status quo.	Guided episodic future thinking; narrative visualisations and scenario tools that strengthen efficacy; participatory futuring that expands Cognitive Alternatives; exercises eliciting anticipated pride/regret for concrete choices; positive visions that increase feasible pathways for change; collective visioning emphasising coordinated societal capacity for action.

Note. The table outlines four leverage points for designing interventions that promote cooperation by reshaping how individuals appraise future collective outcomes, including when each approach is likely to succeed and how it can be operationalised in practice.

Summary statistics of quantitative studies

Figure S11
Lollipop plot of sample sizes of quantitative studies.



Note. Displays the sample size of each study using lollipop plot, with sample size shows on the x-axis. Each point represents an individual study.

Sample size across 133 quantitative studies

- Range: 41 – 12906
- Mean: 759.14
- Median: 378
- SD: 1347.02

Preregistration across 133 quantitative studies

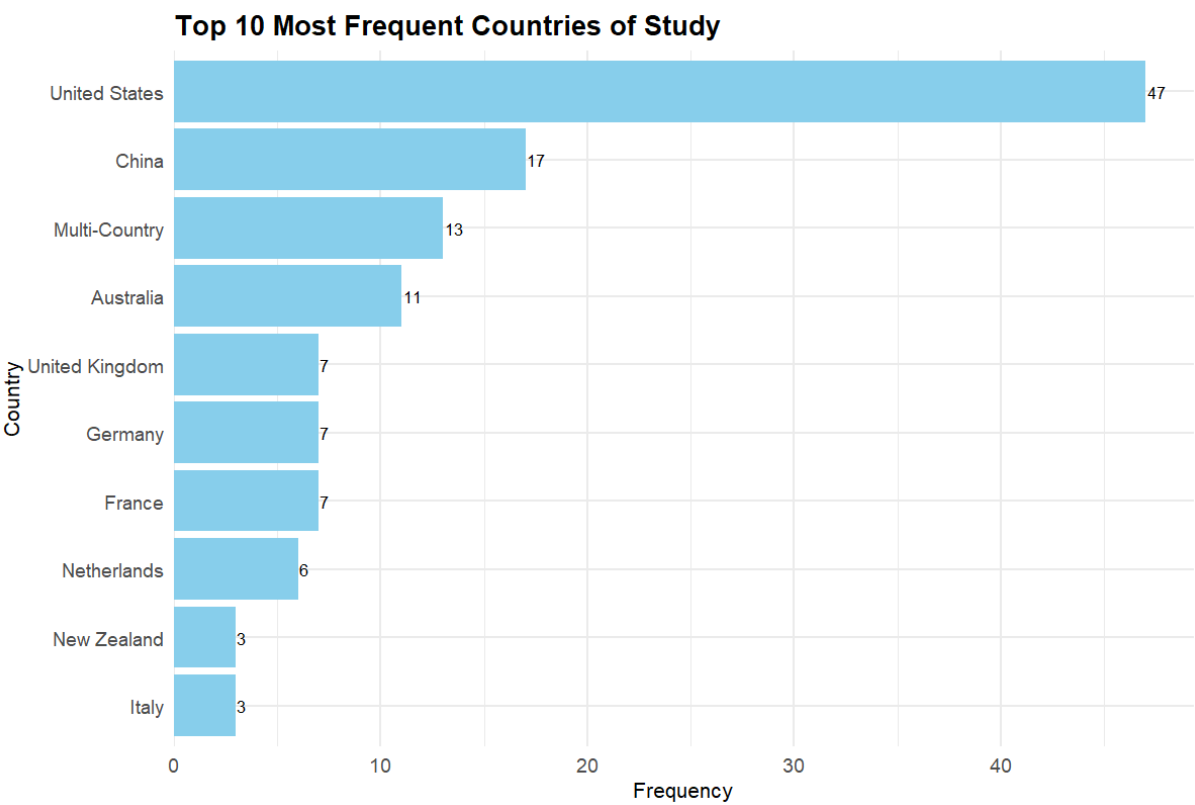
- No 84.2%
- Yes 15.8%

Convenience samples across 133 quantitative studies

- 70.7% (Students, Mturk etc.)

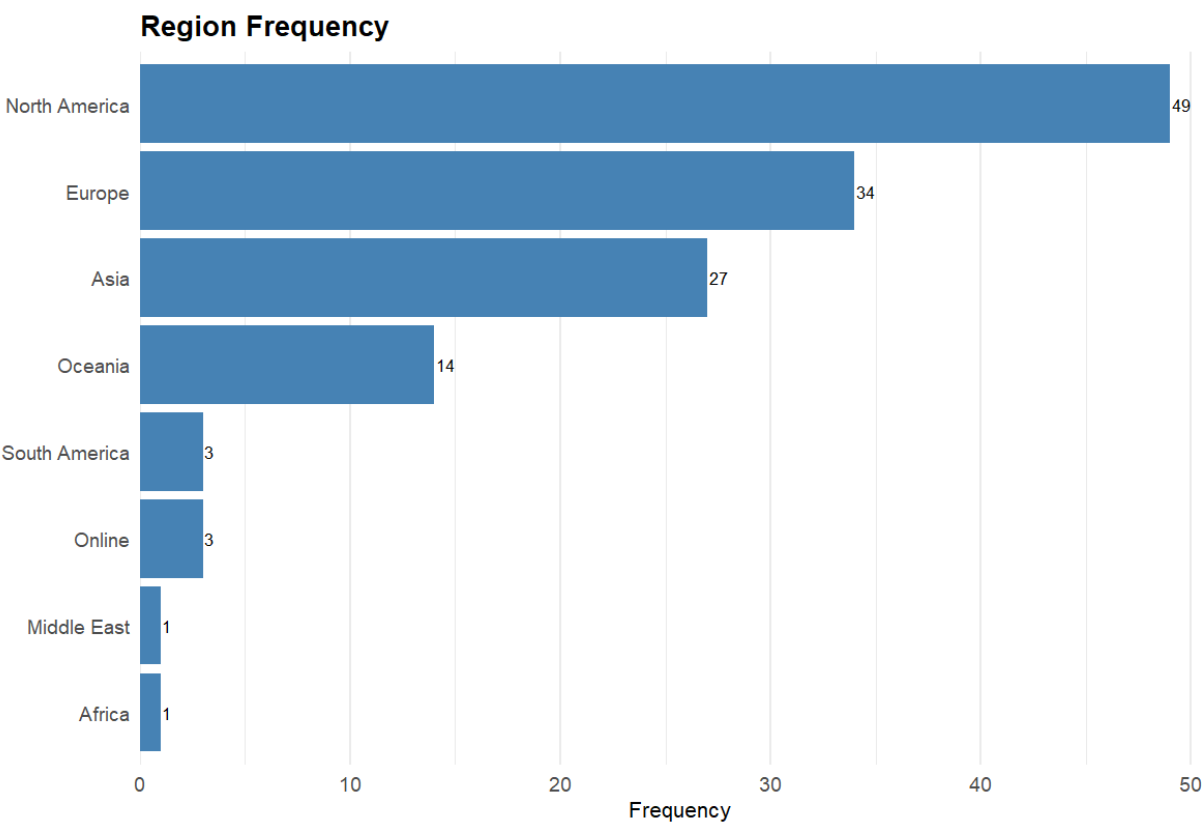
Countries and regions of study participants

Figure S12



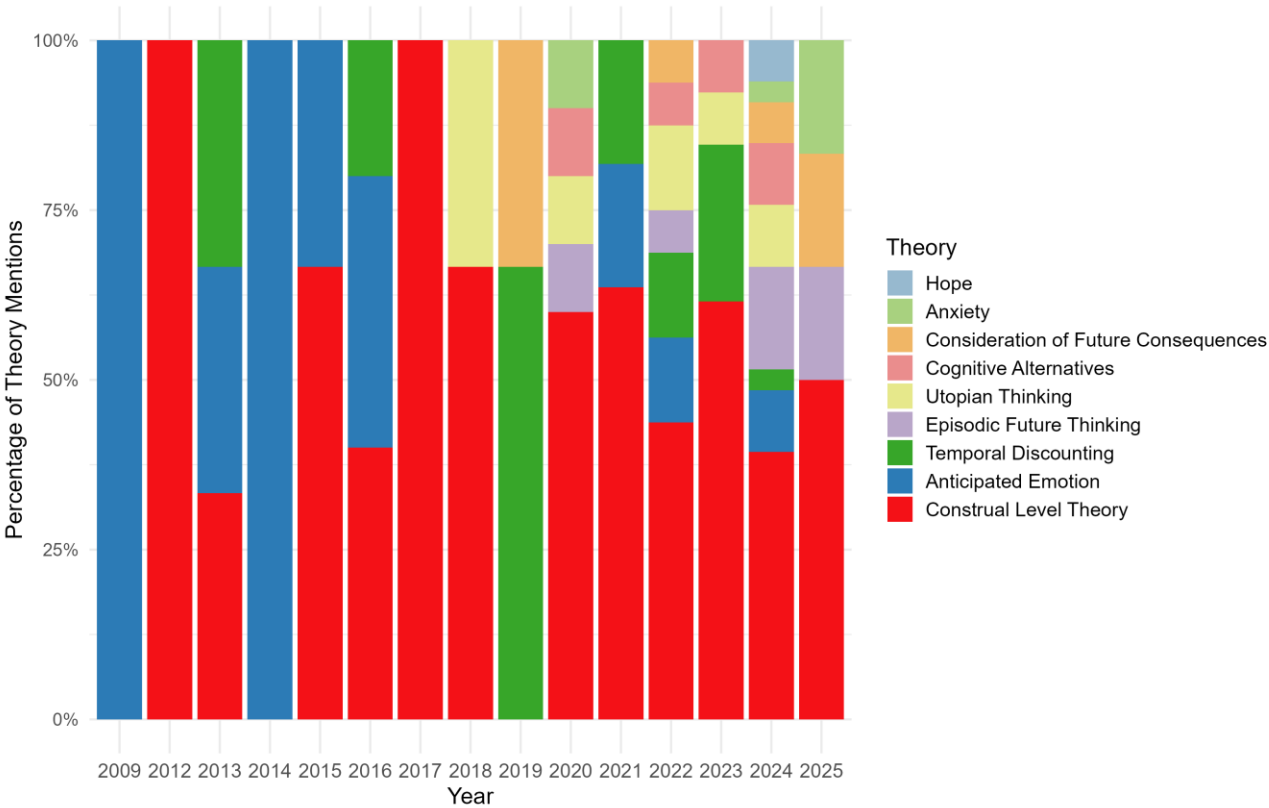
Note. Bar chart of top 10 most frequent countries of study among the quantitative studies.

Figure S13



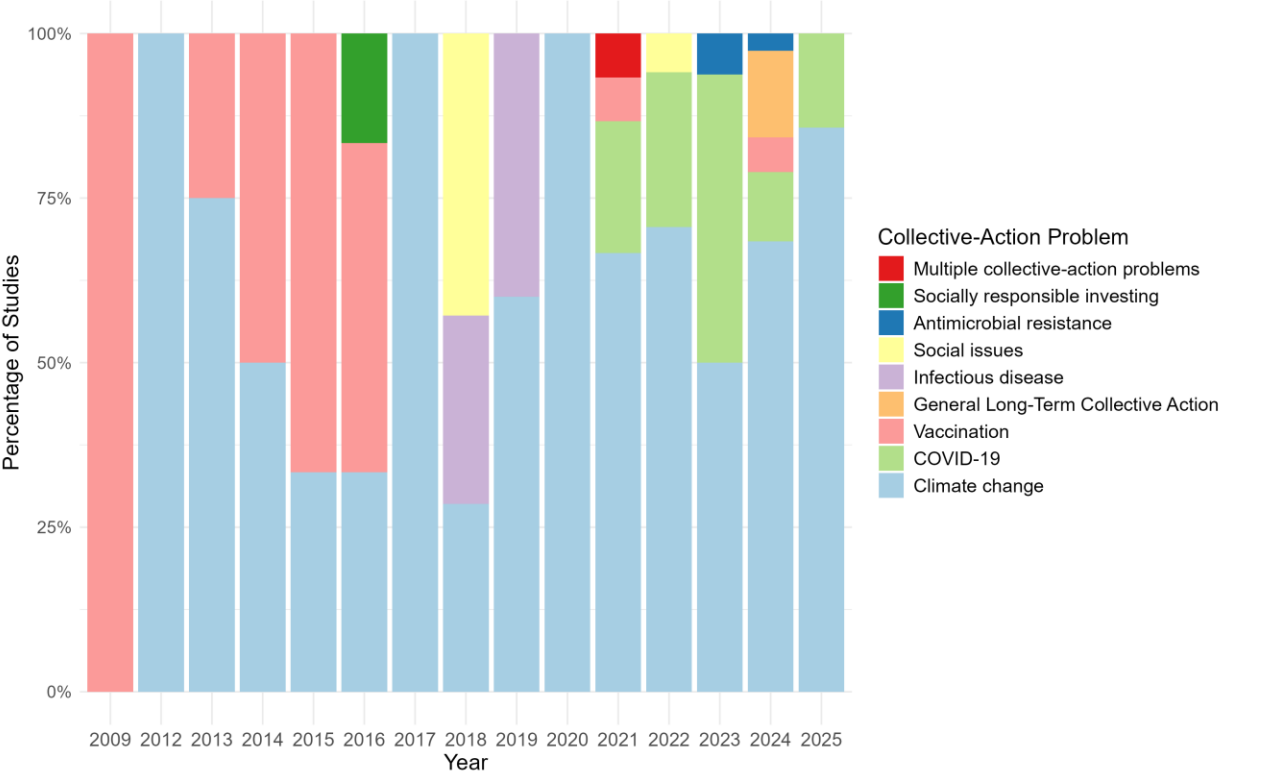
Note. Bar chart of the frequency of each geographic region represented among the quantitative studies.

Figure S14
Proportion of Theories by Year



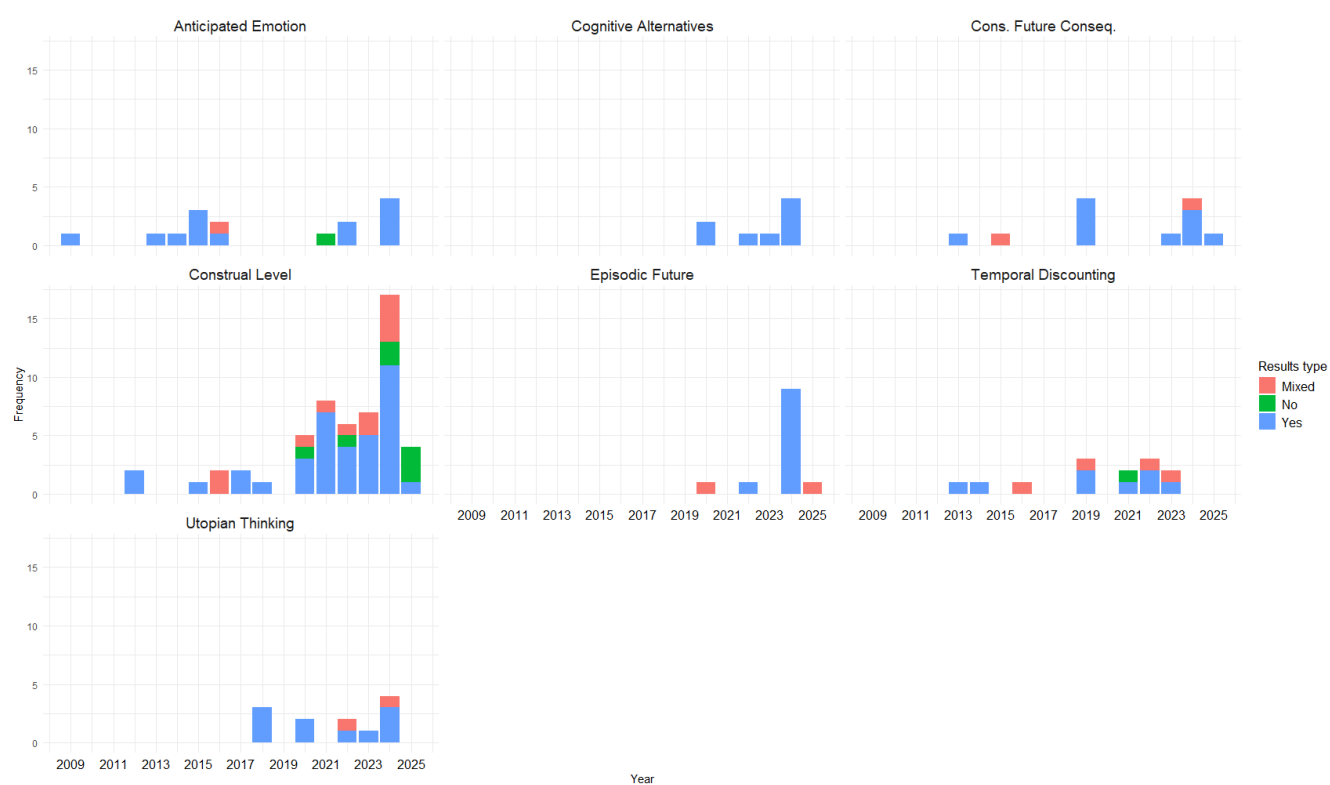
Note. Stacked proportional bar chart showing the distribution of theoretical frameworks across publication years. Each bar represents 100% of studies published in a given year. The x-axis displays publication year and the y-axis displays the percentage of studies.

Figure S15
Proportion of Collective Action Domains by Year



Note. Stacked proportional bar chart showing the distribution of collective-action problem categories across publication years. Each bar represents 100% of studies published in a given year. The x-axis displays publication year and the y-axis displays the percentage of studies.

Figure S16
 Distribution of Study Outcomes by Year



Note. Stacked bar chart showing the proportion of results classified as supportive (blue), non-supportive (green), or mixed (red) with respect to theoretical predictions. The x-axis displays the year of the study’s publication.

Summary of Qualitative Studies

Ahvenharju (2021)

This study identifies three elite groups—progressives, moderates, and sceptics—who share recognition of overconsumption as a problem but diverge sharply in proposed solutions. Progressives favour transformative, systemic policies and exhibit higher futures consciousness (e.g., openness, nature inclusiveness), whereas sceptics prioritise maintaining existing systems and resist disruptive interventions. Moderates occupy an intermediate position, favouring incremental change. Openness to alternative futures emerges as the key differentiator shaping policy preferences, rather than political affiliation alone.

Mansfield (2023)

Through interviews with healthcare worker families, this study shows that adolescent COVID-19 vaccination decisions were shaped by anticipation of future normalcy, safety concerns, and intra-family decision dynamics. Decision-making varied by professional background (physicians more parent-led; nurses more autonomy-oriented), while both parents and adolescents used personal experiences alongside scientific evidence to influence others. Future-oriented motivations (e.g., returning to normal life) and social responsibility played a central role in ultimately high vaccine uptake.

Nguyen et al. (2025)

This study highlights strong links between climate change knowledge, emotional engagement, and behaviour. Most participants reported direct climate impacts, high climate anxiety, and long-term (hyperopic) thinking, which motivated pro-environmental behaviours and sustainable consumption. Emotional drivers—particularly anxiety, guilt, and concern for future generations—interacted with nature connectedness and trust in eco-friendly products, reinforcing long-term decision-making and behavioural change.

Shabnam et al. (2021)

Integrating Construal Level Theory and the Theory of Planned Behaviour, this study shows that psychological distance shapes pro-environmental decisions through a desirability–feasibility trade-off. Distant framings (temporal, social, or abstract knowledge) emphasise values and environmental goals, whereas proximal framings prioritise practical constraints such as cost, availability, and performance. Behavioural intentions depend on how these dimensions interact with attitudes, norms, and perceived behavioural control, highlighting the importance of reducing feasibility barriers to translate future-oriented concern into action.

Wang et al. (2018)

This study reveals stark contrasts between climate scientists and students in emotional engagement and psychological distance. Scientists display strong emotional investment, linking climate change to identity, moral responsibility, and future generations, whereas students adopt a more detached, cognitive framing, often perceiving climate change as distant and deferring responsibility to institutions. Emotional proximity and perceived relevance are key drivers of engagement and action, underscoring the role of affect in reducing psychological distance.

Wang (2025)

Analysing WeChat group discussions during COVID-19, this study shows how collective perceptions and responses evolve through social interaction. Risk perception was high, but attitudes toward control measures varied across groups depending on social norms and group composition. Responses ranged from fear and compliance to scepticism and policy criticism, shifting over time with lived experience and policy change. Collective meaning-making, shaped by communication dynamics and contextual constraints (e.g., censorship), influenced behavioural adaptation and policy evaluation.

Michel-Guillou (2015)

This study demonstrates a disconnect between global climate awareness and local action among water managers. While most participants recognised climate change and its indirect impacts, it was often perceived as temporally and spatially distant, limiting its influence on decision-making. Immediate, local concerns—particularly water quality and pollution—dominated priorities. This highlights how psychological distance constrains the integration of future-oriented risks into practical resource management decisions.

Articles using other theoretical frameworks

Risk Perception Theory

González-Hernández, D. L., Aguirre-Gamboa, R. A., & Meijles, E. W., 2023

DV Mitigation performance (behaviour)

Age (-) → Mitigation performance

Household size (-) → Mitigation performance Household income → Mitigation performance

DV Perceived mitigation efforts (PMEI: intentions / perceived effort)

Environmental concern → PMEI

Belief in climate change → PMEI

Perceived climate change risk → PMEI

Knowledge of how to respond to climate change → PMEI

Gender (woman) → PMEI

Education level → PMEI

Household income → PMEI

Working members → PMEI

Age (-) → PMEI

Household size (-) → PMEI

Household location (urban) → PMEI

Vested Interest Theory

Adame, B. J., & Mattson, L. D., 2025

Behavioural intentions (individual mitigation behaviours)

Salience → Behavioural intentions

Response-efficacy → Behavioural intentions

Gender (woman) → Behavioural intentions (higher)

Policy support (support for climate-friendly candidates)

Salience → Policy support

Immediacy → Policy support

Response-efficacy → Policy support

Gender (woman) → Policy support

Self-Efficacy Theory

Hamann, K. R., Holz, J. R., & Reese, G., 2021

Efficacy beliefs → Sustainability behaviour

Efficacy affect → Sustainability behaviour

Action skills → Sustainability behaviour

Efficacy affect → Private behaviour

Participative efficacy → Volunteering

Action skills → Volunteering

Group identification → Volunteering

Collective efficacy (-) → Volunteering

Collaboration skills (-) → Volunteering

Participative efficacy → Volunteer time

Action skills → Volunteer time

Group identification → Volunteer time

Self-efficacy (-) → Volunteer time

Collaboration skills (-) → Volunteer time

Self-efficacy → Indirect behaviour (persuade or influence others)

Efficacy affect → Indirect behaviour (persuade or influence others)

Action skills → Indirect behaviour (persuade or influence others)

Participative efficacy → Protesting

Action skills → Protesting

Cultural Cognition & Motivated Reasoning

Bain, P. G., Hornsey, M. J., Bongiorno, R., & Jeffries, C., 2012

Study 1 (correlational)

Perceived interpersonal warmth (belief that climate action will make people more caring, considerate, and socially warm) → Environmental citizenship intentions

Perceived societal development (belief that climate action will improve economic growth, technological progress, and scientific advancement) → Environmental citizenship intentions

Study 2 (experimental)

Warmth frame (message emphasising that climate action will make society more caring and considerate) → Environmental citizenship intentions (relative to risk frame)

Development frame (message emphasising that climate action will improve economic and technological progress) → Environmental citizenship intentions (relative to risk frame)

Time Perspective Theory (Zimbardo)/Social Dominance Orientation

Lalot, F., Jauch, M., & Abrams, D., 2022

Right-wing authoritarianism (RWA) (-) → Green agenda support

Ssocial Dominance Orientation (SDO) (-) → Green agenda support

Future thinking → Green agenda support

European identity → Green agenda support

Political orientation (right) (-) → Green agenda support (-)

Political orientation (right) (-) → RWA → Green agenda support

Political orientation (right) (-) → SDO → Green agenda support

Political orientation (right) (-) → Future thinking (-) → Green agenda support

Political orientation (right) (-) → European identity (-) → Green agenda support

Fear-Based Appeal/Utility-based Signal Detection Theory

Sirota, M., & Juanchich, M., 2024

Apocalyptic (post-antibiotic) film (-) → antibiotic expectations (i.e., ↓ belief that one should receive antibiotics for a viral illness)

Apocalyptic (post-antibiotic) film (-) → intentions to request antibiotics (i.e., ↓ stated likelihood of asking a doctor for antibiotics)

Apocalyptic film × COVID-19 experience (-) → desire for antibiotics (effect: COVID-related reduction significant only after viewing the film)

Apocalyptic (post-antibiotic) film (-) → adherence to prescribed antibiotics

Uncertainty Theory

Liu, S., & Mair, J., 2023

Study 1

Uncertainty → controllability → travel intention

Uncertainty → mood state → travel intention

Uncertainty → controllability → mood → travel intention (serial mediation)

Liu, S., & Mair, J., 2023 (cont.)

Study 2

Uncertainty × message framing (gain vs. loss) → travel intention (low uncertainty → gain framing increases intention) (high uncertainty → loss framing increases intention)

Controllability × message framing → travel intention (effect of perceived control depends on gain vs. loss framing)

Uncertainty × message framing × temporal distance → travel intention (framing effects attenuated when travel is in the distant future)

Controllability × message framing × temporal distance → travel intention (conditional interaction within moderated mediation framework)

Uncertainty → controllability → travel intention (indirect effect conditional on message framing)

Uncertainty → controllability → travel intention (indirect effect conditional on temporal distance)

Study 3

Uncertainty × hedonic vs. utilitarian attributes → travel intention (low uncertainty → hedonic increases travel intention) (high uncertainty → utilitarian increases travel intention)

Uncertainty × hedonic vs. utilitarian attributes → mood → travel intention (interaction influences intention through mood)

Uncertainty → mood state → travel intention (indirect effect conditional on hedonic vs. utilitarian framing)

Anxiety + Hyperopia

Nguyen, K. M., Nguyen, N. T., Ngo, T. L., & Le, T. T., 2025

Climate anxiety → Care for environment (Vietnam only)

Climate anxiety → Care for future generations (Vietnam only)

Sustainable trust → Quality of life behaviour (i.e., sustainable consumption motivated by personal wellbeing, health, comfort, and immediate lifestyle benefits)

Sustainable trust → Care for environment

Sustainable trust → Care for future generations

Climate anxiety → Sustainable trust → All behaviours

Climate change risk perception → Sustainable trust → All behaviours

Climate change knowledge → Sustainable trust → All behaviours

Hyperopia × Climate anxiety → Quality of life behaviour (i.e., future orientation strengthens the effect of climate anxiety on wellbeing-oriented sustainable actions; varies by country)

Hyperopia × Climate anxiety → Care for environment (Italy only)

Hyperopia × Sustainable trust → Quality of life behaviour (i.e., future orientation strengthens the translation of trust into wellbeing-oriented sustainable actions; Vietnam only)

Nguyen, K. M., Nguyen, N. T., Ngo, T. L., & Le, T. T., 2025 (cont.)

Hyperopia × Sustainable trust → Care for environment (both countries)

Nature connectedness × Climate risk perception → Sustainable trust (Vietnam only) → Sustainable trust → All behaviours

Anxiety

Salali, G. D., Uysal, M. S., & Bevan, A., 2021

Pandemic-related anxiety → Risk avoidance behaviour (i.e., higher anxiety leads to more protective behaviours such as distancing, masking, avoiding crowds)

Intolerance of uncertainty → Risk avoidance behaviour (i.e., people less tolerant of uncertainty engage more in protective behaviours)

Mindfulness (lower levels) → Risk avoidance behaviour (i.e., lower mindfulness indirectly associated with higher engagement in protective behaviour via anxiety)

Risk perception → Risk avoidance behaviour (Turkey only; small effect) (i.e., perceived infection risk weakly predicts protective behaviour)

Financial satisfaction (lower) → Risk avoidance behaviour (UK only; small effect) (i.e., lower financial satisfaction associated with greater protective behaviour)

Age → Risk avoidance behaviour (i.e., older individuals engage more in protective behaviours)

Sex (female vs male) → Risk avoidance behaviour (i.e., women report higher engagement in protective behaviours)

Education level → Risk avoidance behaviour (context-dependent) (i.e., lower education predicts higher protective behaviour in the UK; mixed effects in Turkey)

Summary of Robustness Check 1

To assess the robustness of the findings, the study outcomes and inductively derived categories were re-analysed across the relevant theoretical frameworks. Overall, only a small proportion of the included studies were preregistered (15.7%), indicating limited use of preregistration practices within this literature. Despite this, all inductive categories were observed at least once across the dataset, suggesting that the identified constructs are consistently represented, albeit unevenly.

However, important imbalances emerged when focusing specifically on preregistered evidence. Notably, none of the studies examining anticipated emotions were preregistered. At the same time, efficacy-related constructs (e.g. vaccine beliefs) were highly prevalent in that literature. As a result, efficacy is comparatively *underrepresented* within the subset of preregistered inductive categories, despite its prominence in the full dataset.

Further variability was observed when examining specific theoretical frameworks. For example, among the 49 articles drawing on Construal Level Theory, only 3 directly were preregistered: one provided supportive evidence, one yielded mixed findings, and one did not support the hypothesised effects. This limited and inconsistent evidence base highlights the need for more rigorous and systematically preregistered tests of theory-driven hypotheses.

Table S23

Outcomes of Preregistered Studies by Theory

Results by Theory (Preregistered Only)								
Counts split by Data Collection Method								
	Survey				Experiments			
	Supportive	Mixed	Non-supportive	Total	Supportive	Mixed	Non-supportive	Total
CA	7 (100%)	0 (0%)	0 (0%)	7	1 (100%)	0 (0%)	0 (0%)	1
CFC	0 (0%)	1 (100%)	0 (0%)	1	2 (100%)	0 (0%)	0 (0%)	2
CLT	0 (0%)	0 (0%)	0 (0%)	0	1 (33%)	1 (33%)	1 (33%)	3
EFT	0 (0%)	0 (0%)	0 (0%)	0	1 (50%)	1 (50%)	0 (0%)	2
TD	1 (100%)	0 (0%)	0 (0%)	1	0 (0%)	0 (0%)	0 (0%)	0
UT	3 (100%)	0 (0%)	0 (0%)	3	1 (50%)	1 (50%)	0 (0%)	2

Note. The table displays outcomes by theoretical frameworks (≥5 papers in the full sample) of preregistered studies only and study design (survey vs experimental), classified as supportive, mixed, or non-supportive of theoretical predictions

Table S24

Construal Level Theory significant predictors, mediators, and moderators by inductive category (preregistered only)

Threat Knowledge	Personal Stake	Efficacy	Emotions	Future Cognition	Identity/ Reputation	Communicative Stimuli	Prosociality	Sociodemographic Factors
risk perception	spatial presence		emotional response		environmental identity	exposure to positive visions virtual reality		

Note. Deduplicated variable representation across inductive categories. Displays all unique variables within significant pathways, and how they were categorised within this scheme.

3/49 articles = Preregistered
 Attanasi et al., 2025; Bosone et al., 2024; Meijers et al., 2023

Table S25

Consideration of Future Consequences: significant predictors, mediators, and moderators by inductive category (preregistered only)

Personal Stake	Emotions	Efficacy	Future Cognition	Identity/ Reputation	Prosociality	Communicative Stimuli	Sociodemographic Factors
disease severity	eco-anxiety		future-focused	reputational concern	empathy induction		men
recent antibiotic use			present-focused (-)		honest-humility		
			future-focused (condition)				
			future-focused (-)				
			cognitive alternatives				

Note. Deduplicated variable representation across inductive categories. Displays all unique variables within significant pathways, and how they were categorised within this scheme.

3/8 articles = Preregistered
Pittaway et al., 2024; Sjästad, 2019; Santana et al., 2023

S26

Temporal Discounting: significant predictors, mediators, moderators, and dependent variables by inductive category (preregistered only)

Threat Knowledge	Personal Stake	Efficacy	Emotions	Future Cognition	Identity/ Reputation	Prosociality	Sociodemographic Factors
	sensitivity to reward magnitude			objective temporal distance (-)		moral obligation	age (-)
				temporal discounting			income (-)

Note. Deduplicated variable representation across inductive categories. Displays all unique variables within significant pathways, and how they were categorised within this scheme.

2/14 articles = Preregistered
Law et al., 2024; Lloyd et al., 2021

Anticipated Emotion (12 articles)—No Preregistered Studies

S27

Cognitive Alternatives: significant predictors, mediators, moderators, and dependent variables by inductive category (unchanged - all studies were preregistered)

Threat Knowledge	Personal Stake	Efficacy	Emotions	Future Cognitions	Identity/ Reputation	Pro sociality	Communicative Stimuli
counterarguing	psychological distance (-)	collective efficacy (-)	eco-anxiety	cognitive alternatives	environmental identity	expansive altruism	cautionary message
		government efficacy	fear	environmental cognitive alternatives	legacy concerns		positive message
		government efficacy (-)	utopian affect	future-focused thinking			
				longtermism			
				present-focused			
				present-focused (-)			
				utopian propensity			
				utopian thinking			

Note. Deduplicated variable representation across inductive categories. Displays all unique variables within significant pathways, and how they were categorised within this scheme.

8/8 articles = Preregistered
Bosone et al., 2024; Pittaway et al., 2024; Wright et al., 2021; Chevrier et al., 2024;
Law et al., 2023; Syropoulos et al., 2024; Troy et al., 2024; Wright et al., 2020

S28

Episodic Future Thinking: significant predictors, mediators, moderators, and dependent variables by inductive category (preregistered only)

Threat Knowledge	Personal Stake	Efficacy	Emotions	Future Cognition	Identity/ Reputation	Pro sociality	Communicative Stimuli	Sociodemographic
				EFT condition		moral obligation		

Note. Deduplicated variable representation across inductive categories. Displays all unique variables within significant pathways, and how they were categorised within this scheme.

2/7 articles = Preregistered
Bø and Wolff, 2020; Law et al., 2024

S29

Utopian Thinking: significant predictors, mediators, moderators, and dependent variables by inductive category (preregistered only)

Threat Knowledge	Personal Stake	Efficacy	Emotions	Future Cognition	Identity/ Reputation	Prosociality	Communicative Stimuli
	psychological distance (-)		fear	antiutopian thinking (-)	environmental identity	expansive altruism	exposure to positive visions
			hope	dystopian thinking		tuiph factor transformative motivation/social justice*	
			hope (-)	environmental cognitive alternatives		tuiph factor* collective orienation/climate resilience	
			tuiph affect*	longtermism			
				tuiph factor economic transformation *			
				tuiph general factor*			
				tuiph propensity*			
				utopian thinking			

Note. Deduplicated variable representation across inductive categories. Displays all unique variables within significant pathways, and how they were categorised within this scheme. TUIPH stands for Transformative Utopian Impulse for Planetary Health and measures individual propensity to think about and engage in addressing social issues pertaining to climate change and human flourishing.

5/8 articles = Preregistered

Bosone et al., 2024; Chevrier et al., 2024; Law et al., 2023; Daysh et al., 2024; Basso and Krpan, 2022

Summary of Robustness Check 2

A follow-up search was conducted on 22 June 2026 to assess the temporal robustness of the synthesis. Searches were performed across available databases (Scopus and PubMed) using the same search string as the primary review. This process identified 16 eligible articles, which were included for descriptive examination.

These studies were analysed using a human–AI assisted approach. For each article, two independent summaries of the results were generated using separate large language model platforms (Microsoft Copilot and DeepSeek). This dual-generation process was used to enhance consistency and reduce platform-specific bias in the interpretation of study findings. To ensure accuracy, a subset of five studies was manually validated by cross-checking the AI-generated summaries against the original results sections of the articles. The extracted summaries and corresponding outcomes were then compared qualitatively with the patterns identified in the primary synthesis.

See Github for the files containing the article titles and summaries under Robustness Check 2.